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Support Needs of Carers Making Proxy Health Care Decisions for People with Dementia; A Systematic Review and Meta-ethnographic Synthesis of Qualitative Studies.

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Abstract

Objective To understand how health care professionals (HCPs) can support carers of people with dementia (PwD) to make proxy health care decisions.

Background There are 50 million dementia sufferers world-wide. Dementia runs a protracted course and comorbidities are common. Decisions about health care often need to be made when the PwD lacks capacity to do so.

Methods A qualitative systematic review protocol was published on PROSPERO. A search was conducted using three online databases (MEDLINE, CINAHL and PsycINFO). The Critical Appraisal Skills Programme (CASP) checklist was used to appraise study quality. A meta-ethnographic synthesis was performed to develop third order constructs.

Study selection Thirteen studies met the inclusion criteria: Primary qualitative research involving carers of PwD who had been involved in making proxy health care decisions.

ResultsThree main domains arose from the meta-ethnography; informational, practical and emotional. Informational support needs included information about dementia itself and the anticipated disease trajectory as well as information on the individual decision to be made. Practical needs included continuity of care, person centred care (PCC) and the use of legal frameworks. Emotional support included recognising the guilt that health care decisions can provoke and the importance of providing guidance in an empathic manner**.**

Conclusions This meta-ethnography highlights opportunities for HCPs and policy makers to improve experiences of carers making proxy health care decisions for PwD.

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Further research into involving carers in PCC could improve the experience of PwD and carers as well as making more appropriate use of health care resources.

Protocol registration The protocol for this systematic review is published in the international prospective register of systematic reviews on 5 June 2020, registration number CRD42020124485. https://www.crd.york.ac.uk/PROSPERO/

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Strengths and Limitations of this study

To our knowledge this is the first systematic review to focus on the experience of proxy health care decision making for carers of people with dementia.

By translating the findings of multiple studies, we make the results more readily available to practitioners, policy makers and researchers.

Although there was a comprehensive search strategy with inclusion of international papers, the study did not include non-English language publications.

Studies may also have been missed due to the variety of terminology used in this field.

Original Protocol

See supplementary file

Funding Statement

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Competing interests

None declared.

Introduction

Dementia affects an estimated 50 million people worldwide with 850,000 sufferers in the UK. As the population ages and deaths from other causes are avoided due to improvements in prevention and treatment, dementia prevalence is expected to rise. The number living with dementia in the UK is predicted to almost double to 1.6 million in the next 10 years.[1] Dementia is the leading cause of death in the UK[2], but the disease often runs a protracted course over many years and most people affected will have other significant health problems. Over 90% of dementia sufferers have at least one comorbidity[1]. Approximately 700,000 people in the UK provide informal care for someone with dementia. It is a particular difficult and stressful condition to care for.[3]

Decisions frequently need to be made about the health care of people with dementia (PwD). These range from decisions to undergo testing to commencing or continuing treatment for dementia or for other conditions. Dementia can be worsened by the disruption to routine and changes to physical health that may result from health care decisions. This must be balanced against the potential gain from accepting care, making these decisions complex. Carers are often required to be involved in proxy decision making about health care as an individual's capacity to make decisions declines.

Shared decision making is a key feature of personalized care which forms part of the NHS long term plan.[4] Shared decision making ensures that individuals are supported to make decisions that are right for them. It is a collaborative process through which a clinician supports a patient to reach a decision about their treatment. The conversation brings together: the clinician's expertise, such as treatment options, evidence, risks and benefits and what the patient knows best: their preferences, personal circumstances, goals, values and beliefs. Shared decision making is known to lead to more realistic expectations, a better match between individuals' values and treatment choices, and fewer unnecessary interventions[5]. The NHS long term plan states that personalized care will underpin care for PwD.[4] Understanding the support needs of carers acting as proxy decision makers will be vital if PwD who have lost decision making capacity and their carers are to benefit from the move towards shared decision making.

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Most published research on making decisions by proxy for PwD has focussed on end of life care[6,7] and nursing home placement.[,8,9,10] Decision making about health care for comorbidities earlier in the disease trajectory has been less well studied. This systematic review and meta-ethnography of qualitative studies aimed to review the existing literature and develop new understanding of the support needs of carers facing health care decisions other than those relating to end of life care and to inform practitioners who may be in a position to provide support.

Method

Patient and Public Involvement

PPI work was conducted with a dementia-specific PPI group based at the University of Sheffield. This shaped an initial scoping literature review and development of the research question.

Search strategy and selection criteria

We searched MEDLINE, PsycINFO, and CINAHL Searches were limited by date from 1st January 1995 to 1st May 2020. Searches were restricted to English language. Qualitative studies and mixed methods studies that clearly reported qualitative findings were eligible for inclusion if they discussed proxy health care decision making.
Target searches using Boolean operators combined terms and Medical Subject Headings (MESH) related to
(1) carer or caregivers
(2) decision making
(3) dementia or alzheimers or cognitive impairment or memory loss

Studies were excluded if they focussed entirely on decisions about nursing home placement or end of life care.

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Quality Appraisal

The qualitative version of the Critical Appraisal Skill Programme (CASP) (Critical Appraisal Skills Programme, 2018)[11] was used to assess methodological quality of the included studies by VH with a sample of papers also being assessed by SL. Discrepancies were resolved by discussion. Quality ratings informed data synthesis and were not used as exclusion criteria.

Data Extraction

Key features of each study were extracted by VH and independently checked by SL and CM. For each study, the following were extracted: author, year of publication, country of origin, stated aim, participant characteristics, method of data collection, qualitative methodology and analytical approach. These features can be found in supplementary table 1.

We followed Noblit and Hare's seven steps of meta-ethnography.[12]
In a meta ethnographic analysis, information is classified as a first, second and third order constructs. First order constructs are direct quotes within studies, second order constructs are authors interpretations within the primary studies, third order constructs are interpretations synthesized by the reviewers from second order constructs.
VH and SL read each paper repeatedly to familiarise themselves before performing line by line coding of all first and second order constructs from the results and discussion sections of each paper. The research team met regularly throughout the review process to discuss the first and second order constructs. These were clustered into candidate third order constructs.
The next stage of the meta-ethnography was to translate the studies into each other. This was done by focussing on each candidate third order construct in turn and compiling lists of first and second order constructs that related to it. Where candidate third order constructs were not robustly supported by the data they were rejected. A process of constant comparison between papers was conducted leading to the

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development of concise statements for each construct (synthesising translations). Both contradictory and directly comparable findings were considered together to develop a line of argument synthesis.

Results

Literature search results

A total of 2019 records were identified from the original database searches. After removing duplicates, initial screening of abstracts and titles, 46 full texts were further assessed for inclusion. At this stage, 33 full texts were excluded, leaving 13 papers.

See figure 1 for PRISMA diagram

Out of 13 studies 5, were conducted in Australia, 5 in the UK, 2 in the USA and 1 in the Netherlands. 9 collected data through interviews only, two used focus groups only, 2 used a combination of focus groups and individual interviews. One of the studies that used interviews, had intended to use focus groups but changed data collection methodology due to difficulty recruiting to focus groups. In total, the included participants across all studies were 184. There was marked variation between the level of information presented about participants. For studies where the information was available, 76% of participants were female and the majority were either spouses/partners or adult children of the person with dementia. Only one study (Livingston) provided data on ethnicity or deprivation. That study had achieved a very diverse sample, however the lack of discussion of diversity in the other studies may indicate that this had either not been sufficiently considered or achieved.

Quality of reporting

CASP ratings ranged from 6 to 10 out of 10, indicating general moderate to high levels of quality in reporting qualitative research. The median of the ratings was 9, with most studies omitting to report reflection on the relationship between the researcher and the participant.

Details of CASP ratings can be found in table 2.

Synthesis

Support needs of carers making proxy health care decisions for PwD fell into 3 main domains; informational support, practical support and emotional support. Third order constructs within this synthesis are illustrated using example first order constructs from the original papers.

Domain 1:

Informational support needs

Carers needed a number of pieces of information to be able to make proxy decisions. This domain captures support needs regarding information on dementia itself as well as details of the health care decision being faced.

Awareness of dementia and disease trajectory

In order to be able to make confident and informed decisions for their loved ones, carers require a sound understanding of dementia, especially with regards to the disease trajectory. An absence of this understanding was a recurring theme in many of the included studies. The reality of inevitable cognitive decline and loss of capacity only became apparent to many participants as they lived through it.

"We didn't realise what dementia meant, the implications. . . I think that people who are carers should receive some training . . . told what to expect and what to do, before it happens, not when it happens." -Livingston

Information about the individual decision to be made

Information content

Shared decision making requires individuals to have sufficient information about the options available and the evidence, risks, benefits for these. In reality, carers described making decisions with incomplete information.

"They didn't say it like it were…we thought there would be one or two [side effects], but I didn't think they'd be as severe as they were"- McWilliams

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"It's also hard to know at each step, is it happening because of the disease or is she medicated? I don't know" -Debellis

Information Delivery

Carers were usually keen to involve the person with dementia as much as possible in decision making. In order to achieve that, information needed to be delivered at a slower than normal pace and carers often went to efforts to translate the information into a more understandable form.

"It's really fast for me…all I could think was well if I wasn't there and you [patient] wouldn't have remembered…for somebody with dementia it needs to be slowed down" -McWilliams

In reality, the PwD was not usually able to make health care decisions due to their complex nature, but the intention to involve them remained important.

"We start off as a joint thing to talk about what needs deciding but normally [it] ends up with me making the final [decision]."-Sinclair

Signposting

Carers often sought informational support from sources other than the HCP. Other carers with experience of making similar decisions were particularly useful.

"I've got an online specialist frontotemporal dementia support forum which is an international forum which consists entirely of other carers and other patients and it's a very knowledgeable forum very well organised very well moderated. And I wish I'd found it 2 years earlier"-Shanley

Appropriate signposting to these services can help carers to feel supported with health care decisions.

Practical Support Needs

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This domain focuses on the practicalities surrounding health care decisions. It deals with how services can be organised to provide better support for carers. It also considers legal support structures.

Continuity of Care

Having a trusted health or social care professional who understood both the patient and the health care system was hugely valued by carers.

''We have a lot of faith in our case manager. He knows the entire dementia network and whatever question or needs we have, he takes care of it.'' -Wolfs

Many carers had not experienced continuity of care but recognized how valuable it could have been when faced with health care decisions.

'when he was initially diagnosed, it would have been great to have somebody from that point onwards that you could have dealt with like one single person or one single number that you could call and say, "This is happening now. What typically happens here? You know, what, what is a good decision to make here?" … So yeah it just would have been good to, to, I suppose just to have some sort of advocacy [Right] for the families just so that you can make informed decisions.'

-Shanley

Person Centred Care

Carers become experts in the PwD as an individual. They act as advocates and their input into health care decisions is essential to achieve true person centred care.

Carers' expertise in their loved one's condition and knowledge of their preferences was routinely overlooked or dismissed by health care staff. This left carers feeling disempowered and frustrated.

"there was never any communication with me, when I did try to say something they just ignored me and listened to him"-Jamieson

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Early planning and legal context

Proxy health care decisions are easier to make when carers have a clear knowledge of the PwDs wishes. Encouraging families to have these discussions early in the disease trajectory whilst the PwD has capacity to do so facilitates future health care decision making.

"Yes, it was very, very difficult. And in hindsight, it would have been a wonderful thing to have had them [conversations] but we didn't."*-Shanley*

These conversations are difficult and some individuals and families need a lot of support with them. Explanation of the potential benefits for carers when they are faced with health care decisions in the future may help some individuals to address this.

"we tried to talk to mummy about what care she would like further down the line, but she never discuss it... she just wouldn't get into that conversation...and I suppose just when we had to start making those decisions mummy was beyond having any input really in it" -Carter

The studies were conducted in a range of countries. Each of the included countries have legal provisions which allow people to formally nominate one or more individuals to act on their behalf should they lack capacity in future. In the UK, this is known as Lasting Power of Attorney (LPA). The studies in this review showed that some participants were aware of the existence of these legal frameworks. A minority had put them in place. Commonly, carers and PwD were resistant to making LPAs.

...I didn't say not at all....I wouldn't do it until much later on I think...I see it as something you hand over when its necessary, not before…-Dening

The mistaken belief that there would be opportunity to set up LPAs once dementia was more advanced led to many reaching a situation where the opportunity was lost.

The only thing that could happen now is Court of Protection . . . because my wife can't sign."

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Carer-Livingston

One technique that some families used to encouraged the PWD to make an LPA was to make them for other family members at the same time. This allowed it to be framed as a generic safeguard in case of future eventualities rather than an individual decision necessitated by their inevitable loss of capacity.

"if we did [lasting power of attorney] for both of them [parents], it wouldn't feel like it was just for my mum because she's dementing."-Livingston

Emotional Support

This domain focusses on the emotional impact of making health care decisions by proxy and how carers can be supported with this.

Guilt

Making proxy health care decisions can be an emotionally fraught process. HCPs need to be mindful of the impact that decision making can have on carers. Checking in with carers about the emotional burden and where necessary signposting to services is important.

"But for a time there, I did feel like I was in charge of ending my mother's life. And I think, if I hadn't had the psychologist to talk that through, I would have found that really difficult, really difficult "-Shanley

The burden of guilt was particularly difficult to manage when carers had to make decisions alone or in conflict with family members.

"My brother and sister . . . wanted the drip, anti-biotics and the oxygen reinstated . . . and the doctor said 'it will flood her heart' and she died a horrific death . . . but my brother always said, 'oh where there's hope' . . . There was no hope." -Livingston

Where decisions were shared, it significantly alleviated the guilt.

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"My case manager explains what options there are and what they mean, and I make the decision together with my mother and my sisters. If you work as a team, you make the best choices.''-Wolfs

Empathic Guidance

When facing proxy decisions, carers appreciated guidance from HCPs. It was important however that those decisions were not made unilaterally by HCPs. Recognition and consideration of the needs and concerns of the PwD and the carer were key. This is approach is described variously in the included papers, but one paper (Shanley) uses the elegant term empathic guidance.

"And, and I think that was, it was very, it was really helpful to have the medical staff kind of put it in, you know, they didn't beat around the bush … I really felt like they were very supportive and they were looking out for her, and they were looking out for me as well. And so I kind of, I did put a lot of trust in what they said … And they were actually considering her needs at the time and, and so they helped me make that decision."-Shanley

Where HCPs did not act as empathic guides and failed to demonstrate consideration of the needs of all parties, attempts to engage carers in decision making were often superficial and unsatisfactory.

"The hospital was no help, and I got aggravated with all of them and finally I had to call a meeting between all the doctors because I could not get anyone to give me any answers."-Truglio Londrigon

Discussion

Summary of main findings

In order to make healthcare decisions for people with dementia, carers require an understanding of the dementia itself and the likely disease trajectory. They also need

information about the decision to be made, presented in a form that can be understood and that can be shared with the PWD wherever possible.

Carers may be facilitated in decision making by access to support groups, especially with carers who have faced similar decisions.

Preparation for future health care decision making should be facilitated early in the dementia trajectory so that carers feel more confident about the PwDs wishes and so that decisions truly are made in their best interests. HCPs should encourage families to formalise arrangements for future decision making using the relevant legal structures such as LPAs.

HCPs who recognize the emotional burden of decision making and act as empathic guides for carers making proxy health care decisions are highly valued.

Strengths and limitations of the study

This systematic review is novel in that it explores proxy health care decision making at earlier stages of the disease trajectory for PwD, rather than a focussing on end of life care or care home placement decision making. We applied rigorous quality assurance to each study and identified strengths and gaps in the evidence base as well as synthesizing key themes of relevance to practice and policy.

A systematic review is by its nature dependent on the quality of the original studies. Of the 13 studies in this review, only one (Livingston) provided detailed demographic data on participants. This study recruited a diverse population in terms of ethnicity, educational background, age and gender. The lack of demographic information in the other studies may suggest a lack of diversity, but certainly suggests a lack of consideration of diversity by the authors. The studies were all conducted in English and excluded the voices of people for whom English is not their first language.

The goal of this review was to focus on the support needs of carers making proxy health care decisions for PwD. Few of the included papers focussed entirely on health care decisions from a carer perspective; many dealt with a range of decision types, others included views of other parties such as the PwD or the HCP. As a result there were often only small sections of the included papers which yielded relevant quotes.

The thematic synthesis was conducted by 4 academic GPs; VH, CM, CH and SL . Whilst every effort was made to maintain a reflexive approach there will undoubtedly have been shared assumptions and blind spots which could have been mitigated with a more diverse research team.

Comparison with existing literature

Information and knowledge about dementia

A lack of knowledge about dementia and the disease trajectory was a common theme throughout this review. A recent systematic review of the public's knowledge and understanding of Alzheimer's disease and dementia found that there was a common misconception that dementia was a normal part of aging.[26] Carers described a need for a post-diagnosis training programme to teach them what to expect. Key within this, was the importance of early advanced care planning. Many participants missed the opportunity to have advanced care planning discussions with their loved ones, in part because they did not realise the inevitable downward trajectory of dementia with resultant loss of capacity. Proxy health care decisions were much easier to make and accept when carers had a clear understanding of their relative's wishes. Supporting carers and PWD to have those conversations early facilitates proxy decision making later on.

Shared decision making

Shared decision making is a key component of the NHS long term plan[4]. It is widely accepted that shared decision making facilitates patients to make decisions that are best for them. A 2011 report by The King's Fund[5] states at the heart of decision making 'is the recognition that clinicians and patients bring different but equally important forms of expertise to the decision-making process'. In the context of proxy-decision making, the carer should take the role of the second expert, however our review suggests that their expertise is commonly disregarded.

These findings echo the Care Quality Commission's 2016 review of how people are involved in their care, which found limited understanding, recording and monitoring of

people's wishes and preferences, inadequate family and carer involvement and insufficient information and explanation of care and support options[27].

Continuity of Care

Many studies have shown the benefits of continuity of care both in terms of patient experience and clinical outcomes.[28,29,30,31]The 2018 NICE dementia guidelines recommend that people living with dementia be provided with a single named health or social care professional who is responsible for coordinating their care and that a program of post diagnostic support be offered to carers.[32] In practice, delivery of this support is inconsistent across the UK, with multiple sectors delivering a range of services with the potential for duplication or gaps.[33] Our review found similarly inconsistent access to care coordination, but where it was available, carers valued it.

Digital Signposting

The emotional burden of decision making for a PWD can be enormous. One of the many consequences of the covid 19 pandemic is the loss of in person support and support networks.[34] Increasing use of on-line support forums offers an alternative for some, but 29% of people aged over 65 in the UK have never used the internet.[35] The movement of services online which has been hugely accelerated by the pandemic risks leaving those without internet access or skills unable to access a wealth of informational, practical and emotional support options.

Implications for future research and clinical practice

The NHS aspires towards making person centred care, including shared decision making part of routine practice. Research is needed to establish how we can achieve this meaningfully for proxy decision makers, so that both they and the PWD have access to the benefits of person centred care.

Continuity of care can be difficult to achieve in the modern health system, but has undoubted benefits for patients with complex comorbidity such as PWD with other health care needs. Commissioners should consider how continuity can best be

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achieved. It should be clear to practitioners and service users who that single point of contact is and how to access them.

In an increasingly digital world, there is potential for widening health inequalities by reducing access to services for those who do not have internet access or computer literacy. A significant minority of older adults do not use the internet and policy makers and commissioners must take measures to ensure that services remain accessible to all, whilst also providing opportunities for people to improve their digital literacy.

In conclusion

Dementia is a large and growing problem. Patients with dementia suffer from inevitable decline in cognition and most will reach a point where they are unable to make health care decisions independently. If PWD are to benefit from the drive towards patient centred care, it is vital that we consider the support needs of their carers.

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Data Statement No additional data are available

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Table 1: Overview of Study Characteristics

Author, year and country	Stated aim	Participant characteristics	Method of data collection	Analytic approach
Bloomer et al 2016 Australia	Explore the experiences of family carers of people with dementia who are hospitalised	20 carers age range 34-92 equal gender distribution mixture of spouses, adult children and friends (no details provided)	Individual face to face interviews	Thematic analysis
Shanley et al 2017 Australia	Providing support to surrogate decision-makers for people living with dementia: Healthcare professional, organisational and community responsibilities	34 surrogate decision makers. 79 % female 85% adult children mean age 57.4	Individual face to face or telephone interviews	Thematic analysis
Sinclair et al 2018 Australia	Explore how couples with dementia experience healthcare, lifestyle, and everyday decision-making	15 carers Aged 59-82 60% female All spouse partners	Face to face interviews, 6 individual, 9 couple dyad interviews	Thematic analysis
Truglio-Londrigan 2018 USA	To identify Alzheimers Disease specific decisions that past and present caregivers faced over the disease trajectory	13 carers 85% female 9 adult children, 3 spouses	Focus groups	Thematic analysis
DeBellis 2017 Australia	To explore relatives' experience, knowledge and perceptions of challenging behavioural and psychological symptoms of dementia and	6 participants, all F age 45-62, 4 daughters, one wife, one daughter in law	Face to face interviews	Thematic analysis

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	association with antipsychotic use for persons with dementia in residential aged care.			
McWilliams 2018 UK	To investigate cancer-related information needs and treatment decision making experiences of people with cancer-dementia, their informal caregives and oncology HCPs	9 carers mean age 73.6 44% female 56% spouse, 44% child	Face to face interviews	Thematic analysis
Dening 2017 UK	To understand the lived context of healthcare decision making in the past, present and future for people with dementia and their family carer	6 carers age 49-85 5 spouses, 1 adult child	Face to face interviews	Thematic analysis
Elliot et al 2007 USA	Investigating the decision making of family members of institutionalised older adults with advanced cognitive impairment, with particular focus on spirituality and meaning	39 family members average age 62 4 same generation (spouse/sibling) 35 adult children	Focus groups	Constant comparative method
Carter 2018 UK	Exploring the experiences and preparedness of family carers for best interest decision-making of a relative living with advanced dementia	20 carers 75% female 4 siblings, 2 spouse, 13 children, 1 extended family member	Face to face interviews	Thematic analysis
Livingston 2010	Exploring how carers make decisions for people	Focus groups: 43 participants Individual interviews: 46	Focus groups to generate list of difficult decisions,	Grounded theory

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UK	with dementia who lack capacity		Individual interviews to explore in depth	
Walker 2001 UK	Investigating how we facilitate carer's involvement in decision making?	20 carers of people with dementia No further details on participants provided	Face to face individual interviews	Constant comparative method
Wolfs 2012 The Netherlands	Gaining carer's insights into the decision making process about treatment and care in dementia patients	26 carers 50% female 11 spouses, 12 adult children,3 other	Brief individual interviews inform interview schedule, then focus groups	Grounded theory
Jamieson 2016 Australia	Investigating the experience of people with dementia and their carers when transitioning from home to hospital	30 carers 87% female	Started with focus groups, but as these were difficult for carers to access, changed to individual telephone interviews	Thematic analysis

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	Bloomer	Carter	DeBellis	Dening	Elliott	Jamieson	Livingston	McWilliams	Shanley	Sinclair	Trugliio-Londrigon	Walker	Wolfs
Was there a clear statement of aims of the research?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes
Is qualitative methodology appropriate?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the research design appropriate to address the aims of the research?	yes	yes	no	yes	no	yes	yes	yes	yes	yes	yes	yes	yes
Was the recruitment strategy appropriate to the aims of the	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes

research													
Was the data collected in a way that addressed the research issue?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes
Has the relationship between researcher and participants been adequately considered?	no	no	no	no	no	no	no	no	no	no	no	no	yes
Have ethical issues been taken into consideration?	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data analysis sufficiently rigorous?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Is there a clear statement of findings?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
How valuable is the research?	high	high	medium	medium	low	high	high	high	high	high	high	moderate	high
	9	9	8	8	6	9	9	9	9	9	9	8	10

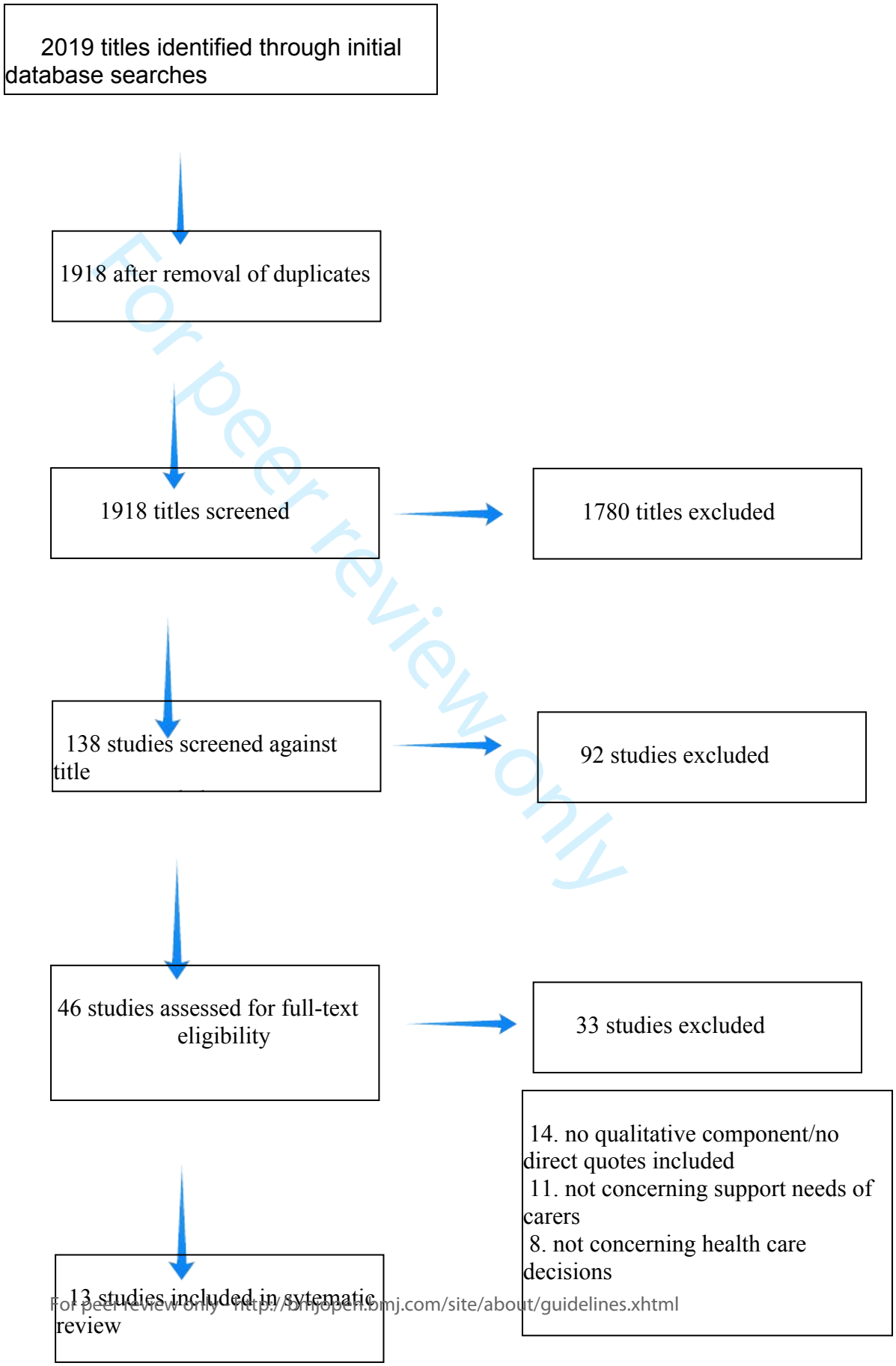
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Table 2: CASP ratings

	Bloomer	Carter	DeBellis	Dening	Elliott	Jamieson	Livingston	McWilliams	Shanley	Sinclair	Truglio-Londrigon	Walker	Wolfs
Was there a clear statement of aims of the research?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes
Is qualitative methodology appropriate?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the research design appropriate to address the aims of the research?	yes	yes	no	yes	no	yes	yes	yes	yes	yes	yes	yes	yes
Was the recruitment strategy appropriate to the aims of the research	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data collected in a way that addressed the research issue?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes
Has the relationship between researcher and participants been adequately considered?	no	no	no	no	no	no	no	no	no	no	no	no	yes
Have ethical issues been taken into consideration?	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data analysis sufficiently rigorous?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Is there a clear statement of findings?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
How valuable is the research?	high	high	medium	medium	low	high	high	high	high	high	high	medium	high
CASP rating	9	9	8	8	6	9	9	9	9	9	9	8	10

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Figure 1: PRISMA diagram



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NHS National Institute for Health Research

UNIVERSITY of York
Centre for Reviews and Dissemination

Systematic review

1. * Review title.

Give the title of the review in English

What are the support needs of family carers making proxy decisions about healthcare treatment for a relative with dementia?

2. Original language title.

For reviews in languages other than English, give the title in the original language. This will be displayed with the English language title.

3. * Anticipated or actual start date.

Give the date the systematic review started or is expected to start.

02/06/2020

4. * Anticipated completion date.

Give the date by which the review is expected to be completed.

08/12/2020

5. * Stage of review at time of this submission.

Tick the boxes to show which review tasks have been started and which have been completed. Update this field each time any amendments are made to a published record.

Reviews that have started data extraction (at the time of initial submission) are not eligible for inclusion in PROSPERO. If there is later evidence that incorrect status and/or completion date has been supplied, the published PROSPERO record will be marked as retracted.

This field uses answers to initial screening questions. It cannot be edited until after registration.

The review has not yet started: No

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Review stage	Started	Completed
Preliminary searches	Yes	Yes
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

Provide any other relevant information about the stage of the review here.

6. * Named contact.

The named contact is the guarantor for the accuracy of the information in the register record. This may be any member of the review team.

Victoria Hodges

Email salutation (e.g. "Dr Smith" or "Joanne") for correspondence:

Dr Hodges

7. * Named contact email.

Give the electronic email address of the named contact.

v.hodges@sheffield.ac.uk

8. Named contact address

Give the full institutional/organisational postal address for the named contact.

Academic Unit of Primary Medical Care, University of Sheffield Sam Fox House Northern General Hospital

Herries Road Sheffield S5 7AU

9. Named contact phone number.

Give the telephone number for the named contact, including international dialling code.

+44 (0)7886963388

10. * Organisational affiliation of the review.

Full title of the organisational affiliations for this review and website address if available. This field may be completed as 'None' if the review is not affiliated to any organisation.

University of Sheffield

Organisation web address:

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Health Research

11. * Review team members and their organisational affiliations.

Give the personal details and the organisational affiliations of each member of the review team. Affiliation refers to groups or organisations to which review team members belong. **NOTE: email and country now MUST be entered for each person, unless you are amending a published record.**

Dr Catherine Hynes. University of Sheffield
Dr Rob Heminway. Sheffield Health and Social Care Trust
Dr Caroline Mitchell. University of Sheffield
Professor Lynda Wyld. University of Sheffield
Dr Victoria Hodges. University of Sheffield

12. * Funding sources/sponsors.

Details of the individuals, organizations, groups, companies or other legal entities who have funded or sponsored the review.

RCGP SFB 2018-07

Grant number(s)

State the funder, grant or award number and the date of award

RCGP SFB 2018-07

13. * Conflicts of interest.

List actual or perceived conflicts of interest (financial or academic).

None

14. Collaborators.

Give the name and affiliation of any individuals or organisations who are working on the review but who are not listed as review team members. **NOTE: email and country must be completed for each person, unless you are amending a published record.**

15. * Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

a) Describe family carer experiences of any aspect of decision making about healthcare treatment for a relative with dementia.

b) Identify barriers and facilitators family carers experience when making proxy decisions about healthcare treatment for a relative with dementia.

c) Describe interventions which have been developed to aid family members make proxy decisions about healthcare treatment for a relative with dementia.

P – family carers of people with dementia

I – proxy decision making about a healthcare treatment

C – n/a

O – support needs

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16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

Electronic databases: MEDLINE, CINAHL, PsycINFO, Embase; Google search using same search terms, limiting to web pages where search terms appear in the summary; ClinicalTrials.gov

Exclusion criteria:

a) Published prior to 1995

b) Non-English language

c) Only report proxy decisions not related to healthcare

d) Reviews, editorials, dissertations, meeting abstracts and correspondence

17. URL to search strategy.

Upload a file with your search strategy, or an example of a search strategy for a specific database, (including the keywords) in pdf or word format. In doing so you are consenting to the file being made publicly accessible. Or provide a URL or link to the strategy. Do NOT provide links to your search **results**.

https://www.crd.york.ac.uk/PROSPEROFILES/124485_STRATEGY_20190205.pdf

Alternatively, upload your search strategy to CRD in pdf format. Please note that by doing so you are consenting to the file being made publicly accessible.

Do not make this file publicly available until the review is complete

18. * Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

Family/informal carers of people with dementia.

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

People of any age and gender who have a close relative with dementia (regardless of where their relative resides).

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The preferred format includes details of both inclusion and exclusion criteria.

Proxy-decision making about a healthcare treatment (of any form).

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

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Not applicable.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Qualitative, mixed methods.

23. Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

Studies exploring family carer experiences of making decisions about healthcare treatment on behalf of a relative with dementia.

Any form of healthcare treatment

Not decisions relating to something other than healthcare treatment, for example care home placement

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurement are made, if these are part of the review inclusion criteria.

Experiences, support needs, barriers, facilitators of family/informal carers making proxy decisions about halthcare treatment for a relative with dementia

Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate to the review

None.

Measures of effect

Please specify the effect measure(s) for you additional outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not applicable.

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

Screening PRISMA guidelines. Independent researchers (VH, CH, RH and CM) will screen results of the initial searches by title +/- abstract to exclude irrelevant studies

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Eligibility – full text articles will then be assessed for eligibility

The following data will be extracted from eligible papers:

Study details: location/country, research question/objectives

Population: Age, Gender, Ethnicity

Recruitment sampling: Inclusion criteria, response rate, test of any differences between those who responded and those who did not.

Intervention:Proxy-decision making about a healthcare treatment (of any form)

Outcome: Experiences, support needs, barriers, facilitators of family/informal carers making proxy decisions about healthcare treatment for a relative with dementia

Data Collection Method: eg survey, focus groups, interviews

Validation and recording: eg validation of survey tools, interview transcription

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

Two independent researchers will conduct validity assessments for each paper selected for inclusion:

COREQ for qualitative studies

MMAT for mixed methods

Assessment will be done at study level

Discrepancies will be resolved by discussion and if necessary by involving a third reviewer.

A discussion of the validity of each included study will be included in the final report

28. * Strategy for data synthesis.

Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta-analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.

A narrative synthesis will be produced. Nvivo softwear will be used to aid the team to code conceptual findings as they appear in the paper and to allow the team to compare interpretations. Statistical methods will not be employed. Included papers will be analysed for key themes in relation to the support needs expressed by family carers at the time of proxy decision making about healthcare treatment for a relative with dementia. Primary analysis will be done by CH and RH and discussed with CM, LW. Limitations of the review will be considered.

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

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None planned.

30. * Type and method of review.

Select the type of review, review method and health area from the lists below.

Type of review

Cost effectiveness
No

Diagnostic
No

Epidemiologic
No

Individual patient data (IPD) meta-analysis
No

Intervention
No

Meta-analysis
No

Methodology
No

Narrative synthesis
Yes

Network meta-analysis
No

Pre-clinical
No

Prevention
No

Prognostic
No

Prospective meta-analysis (PMA)
No

Review of reviews
No

Service delivery
No

Synthesis of qualitative studies
Yes

Systematic review
Yes

Other
No

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Health area of the review

Alcohol/substance misuse/abuse
No

Blood and immune system
No

Cancer
No

Cardiovascular
No

Care of the elderly
Yes

Child health
No

Complementary therapies
No

COVID-19
No

Crime and justice
No

Dental
No

Digestive system
No

Ear, nose and throat
No

Education
No

Endocrine and metabolic disorders
No

Eye disorders
No

General interest
No

Genetics
No

Health inequalities/health equity
No

Infections and infestations
No

International development
No

Mental health and behavioural conditions
Yes

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Musculoskeletal
No

Neurological
No

Nursing
No

Obstetrics and gynaecology
No

Oral health
No

Palliative care
No

Perioperative care
No

Physiotherapy
No

Pregnancy and childbirth
No

Public health (including social determinants of health)
No

Rehabilitation
No

Respiratory disorders
No

Service delivery
No

Skin disorders
No

Social care
No

Surgery
No

Tropical Medicine
No

Urological
No

Wounds, injuries and accidents
No

Violence and abuse
No

31. Language.

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Select each language individually to add it to the list below, use the bin icon to remove any added in error.
English

There is not an English language summary

32. * Country.

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If the protocol for this review is published provide details (authors, title and journal details, preferably in Vancouver format)

Add web link to the published protocol.

Or, upload your published protocol here in pdf format. Note that the upload will be publicly accessible.

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Do you intend to publish the review on completion?

Yes

Give brief details of plans for communicating review findings.?

It will be used to inform the interview topic guide for a qualitative study funded by the RCGP: Cancer and dementia: carer experiences of proxy decision making (IRAS 252671, CMPS 40597)

36. Keywords.

Give words or phrases that best describe the review. Separate keywords with a semicolon or new line. Keywords help PROSPERO users find your review (keywords do not appear in the public record but are included in searches). Be as specific and precise as possible. Avoid acronyms and abbreviations unless these are in wide use.

Dementia; treatment; decision-making; proxy; caregivers

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Update review status when the review is completed and when it is published.New registrations must be ongoing so this field is not editable for initial submission.
Please provide anticipated publication date

Review_Ongoing

39. Any additional information.

Provide any other information relevant to the registration of this review.

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Give the link to the published review or preprint.

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PRISMA 2009 Checklist (Adapted for KIN 4400)

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a literature review.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings;	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known about your topic.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Eligibility criteria	5	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	
Information sources	6	Describe all information sources (e.g., databases with dates of coverage) in the search and date last searched.	6
Search	7	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	6
Study selection	8	State the process for selecting studies (i.e., screening, eligibility).	6
Risk of bias in individual studies	9	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level).	8
Risk of bias across studies	10	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	3
RESULTS			
Study selection	11	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7

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PRISMA 2009 Checklist (Adapted for KIN 4400)

Section/topic	#	Checklist item	Reported on page #
Study characteristics	12	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6
Synthesis of results of individual studies	13	For all outcomes considered (benefits or harms), present, for each study: (a) summary of results and (b) relationship to other studies under review (e.g. agreements or disagreements in methods, sampling, data collection or findings).	7
DISCUSSION			
Summary of evidence	14	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	15
Limitations	15	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
CONCLUSION			
Conclusions	16	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	17

Adapted from: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA statement. *PLoS Medicine*, 6(6), e1000097. doi:10.1371/journal.pmed1000097

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Support Needs of Carers Making Proxy Healthcare Decisions for People with dementia; a systematic review based on the Noblit and Hare meta-ethnographic synthesis of qualitative studies.

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Support Needs of Carers Making Proxy Healthcare Decisions for People with dementia; a systematic review based on the Noblit and Hare meta-ethnographicsynthesis of qualitative studies.

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Key Words: Dementia; Caregivers; Decision making, shared; Advanced care planning

Word count: 4583

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Abstract

Background There are 50 million dementia sufferers worldwide. Decisions about healthcareoften need to be made when the person with dementia lacks capacity to do so. Understanding the support needs of carers acting as proxy healthcare decision makers will be vital in improving the decision making process for people with dementia and addressingthe holistic needs of carers.

Objective The objective of this study was to review the existing literature on the supportneeds of carers acting as proxy healthcare decision makers for people with dementia.

Design A qualitative systematic review protocol was published on PROSPERO. The CriticalAppraisal Skills Programme (CASP) checklist was used to appraise study quality. A meta-ethnographic synthesis was performed to develop third order constructs.

Data Sources A search was conducted using three online databases (MEDLINE, CINAHLand PsycINFO).

Eligibility criteria Fifteen studies met the inclusion criteria: Primary qualitative research involving carers of people with dementia who had been involved in making proxy healthcaredecisions.

Data extraction and synthesis Two independent researchers conducted validity assessments for each paper selected for inclusion and discrepancies were resolved by discussions with a third reviewer. Nvivo software was used and conceptual findings fromstudy papers lead to interpretations of findings by the team.

Results From the 15 papers included in the study three main domains arose from the meta-ethnography; informational, practical and emotional. Informational support needs included information about dementia itself and the anticipated disease trajectory. Practicalneeds included continuity of care, person centred care and the use of legal frameworks.

Emotional support included recognising the guilt that healthcare decisions can provoke andthe importance of providing guidance in an empathic manner**.**

Conclusions This meta-ethnography highlights opportunities for health care professionals and policy makers to improve experiences of carers making proxy healthcare decisions forpeople with dementia.

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Strengths and Limitations of this study

- To our knowledge this is the first systematic review to focus on the experience ofproxy healthcare decision making for carers of people with dementia.
- By translating results of multiple studies, we make the results more readily availableto practitioners, policy makers and researchers.
- Although there was a comprehensive search strategy with inclusion of internationalpapers, the study did not include non-English language publications.
- Studies may also have been missed due to the variety of terminology used in thisfield.

Funding Statement

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Competing interests

None declared.

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Introduction

Dementia affects an estimated 50 million people worldwide with 850,000 sufferers in the UK. As the UK population ages and deaths from other causes are avoided due to improvements inprevention and treatment, dementia prevalence is likely to rise. The number living with dementia in the UK is expected to almost double to 1.6 million in the next 10 years.[1] Dementiais the leading cause of death in the UK,[2] but the disease often

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runs a protracted course over many years and most people affected will have other significant health problems. Over 90% of dementia sufferers have at least one comorbidity.[1] Approximately 700,000 people in the UK provide informal care for someone with dementia. It is a particular difficult and stressfulcondition to care for.[3]

Decisions frequently need to be made about the healthcare of people with dementia. These range from decisions to undergo testing to commencing or continuing treatment for dementiaor for other conditions. Dementia can be worsened by the disruption to routine and changes to physical health that may result from healthcare decisions. This must be balanced against the potential gain from accepting care making these decisions complex. Carers are often required to be involved in proxy decision making about healthcare as an individual's capacityto make decisions declines.

Shared decision making is a key feature of personalized care which forms part of the NHS long term plan.[4] Shared decision making ensures that individuals are supported to make decisions that are right for them. It is a collaborative process through which a clinician supports a patient to reach a decision about their treatment. The conversation brings together: the clinician's expertise, such as treatment options, evidence, risks and benefits andwhat the patient knows best: their preferences, personal circumstances, goals, values and beliefs. Shared decision making is known to lead to more realistic expectations, a better match between individuals' values and treatment choices, and fewer unnecessary interventions.[5] The NHS long term plan states that personalized care will underpin care for people with dementia.[4] Understanding the support needs of carers acting as proxy decision makers will be vital if people with dementia who have lost decision making capacity and their carers are to benefit from the move towards shared decision making. The rationale for this systematic review and meta-ethnography of qualitative studies was to review the existing literature with a view to developing new understanding of the support needs of carers for people with dementia facing healthcare decisions and to inform practitioners who may be in a position to provide support. Most published research on makingdecisions by proxy for people with dementia has focused on end of life care [6-7] and nursing home placement. [8-10] Decision making about healthcare for comorbidities earlier in the disease trajectory has been less well studied.

Research Objectives

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- Describe family carer experiences of proxy decision making about healthcare treatment fora relative with dementia excluding decisions about end of life care.
- Identify barriers and facilitators family carers experience when making proxy decisionsabout healthcare treatment for a relative with dementia.

Method

Patient and Public Involvement

PPI work was conducted with a dementia-specific PPI group based at the University of Sheffield. This shaped an initial scoping literature review and development of the researchquestion.

Eligibility criteria

Inclusion criteria: a) People of any age and gender who have a close relative with dementia(regardless of where they reside), b) Proxy-decision making about a healthcare treatment

- Any primary, empirical researchQualitative, mixed methods

Exclusion criteria: a) Published prior to 1995 b) Non-English language c) Only report proxy decisions not related to healthcare d) End of Life e) Reviews, editorials, dissertations, meetingabstracts and correspondence

Information sources

We searched MEDLINE, PsycINFO, and CINAHL Searches were limited by date from 1st January 1995 to 1st May 2020. Searches were restricted to English language. Searches wererepeated on the 22nd June 2021.

Search strategy and selection criteria

The objective of this study was to review the existing literature on the support needs of carersacting as proxy healthcare decision makers for people with dementia. The protocol for this systematic review was published in the international prospective register of systematic reviews on 5 June 2020, registration number CRD42020124485. https://www.crd.york.ac.uk/PROSPERO

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Qualitative studies that clearly reported qualitative findings were eligible for inclusion if theydiscussed proxy healthcare decision making.

Target searches using Boolean operators combined terms and Medical Subject Headings(MESH) related to

- carer or caregiver or informal carer or family carer or relative AND
- Decision making or Proxy or Decide or Surrogate decision mak* or 3 rd party decisionmak* AND
- dementia or alzheimers AND
- Qualitative

Study selection Studies were excluded if they focused entirely on decisions about nursing orhome placement or end of life care and the steps of study selection are represented in the PRISMA diagram.

Quality Appraisal

The qualitative version of the Critical Appraisal Skill Programme (CASP) (Critical Appraisal Skills Programme, 2018) [11] was used to assess methodological quality of the included studies by VH with a sample of papers also being assessed by SL. Discrepancies were resolved by discussion. Quality ratings informed data synthesis and were not used as exclusion criteria. The CASP was also used to reduce the risk of reporting bias and discussions of the validity ofeach study was done during data extraction and synthesis.

Data Extraction

Key features of each study were extracted by VH and independently checked by SL and CM.For each study, the following were extracted: author, year of publication, country of origin, stated aim, participant characteristics, method of data collection, qualitative methodology andanalytical approach. The outcomes that were sought in the papers are summarized in Table 1and these features can be found in the supplementary Table 1.

Table 1: PICO

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Sample	People of any age and gender who have a close relative with dementia (regardless of where they reside)
Phenomenon of interest	Proxy-decision making about a healthcare treatment
Design	Any qualitative primary, empirical research
Evaluation (Outcomes)	Experiences, support needs, barriers, facilitators of family/informal carers making proxy decisions about healthcare treatment for a relative with dementia

We followed Noblit and Hare's seven steps of meta-ethnography.[12] In a meta ethnographicanalysis, information is classified as a first, second and third order constructs. First order constructs are direct quotes within studies, second order constructs are authors' interpretations within the primary studies, third order constructs are interpretations synthesized by the reviewers from second order constructs.

VH and SL read each paper repeatedly to familiarise themselves before performing line by line coding of all first and second order constructs from the results and discussion sections of each paper. The research team met regularly throughout the review process to discuss the first and second order constructs. These were clustered into candidate third order constructs. The next stage of the meta-ethnography was to translate the studies into each other. This wasdone by focusing on each candidate third order construct in turn and compiling lists of first and second order constructs that related to it. Where candidate third order constructs were not robustly supported by the data they were rejected. A process of constant comparison between papers was conducted leading to the development of concise statements for each construct (synthesising translations). Both contradictory and directly comparable findings were considered together to develop a line of argument synthesis. An example of how this was applied in this study is seen in this table below.

Table 2

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Construct	Explanation	Example
First order constructs	Carers accounts of their experiences	*'it's a lot of negotiation… sometimes…it's a case of having a familiar face'* *I think, you know, when he was initially diagnosed, it would have been great to have somebody from that point onwards that you could have dealt with like one single person or one single number that you could call and say, "This is happening now. What typically happens here? You know, what, what is a good decision to make here?" … So yeah it just would have been good to, to, I suppose just to have some sort of advocacy [Right] for the families just so that you can make informed decisions.*
Second order constructs	Author's views and interpretations expressed as themes in the primary papers	Similarly, healthcare professionals unfamiliar with the family and the resident's individual wishes were also noted to cause unnecessary anxiety, again resulting in reluctance of further contact. The experience of an uncoordinated discharge process was for family and friends compounded by their own exhaustion after their hospital vigil. The tiredness was further complicated by delays in restarting community services that were accessed previously This health professional was often their GP. Having this consistent, ongoing support helped the surrogate decision- maker become aware of issues and find their way through the complex maze of service provision. It meant that surrogate decision- makers did not have to keep repeating their story and building trust with new professionals.
Third order constructs	The views and interpretations of the synthesis team	Continuity of Care Having a trusted health or social care professional who understood both the patient and the healthcare system was hugely valued by carers.

Bias assessment

Two independent researchers conducted validity assessments for each paper selected for inclusion and discrepancies were resolved by discussions with a third reviewer. Nvivo software was used and conceptual findings from study papers lead to interpretations of findings by the team. A thorough CASP for qualitative studies was done to reduce for otherbiases (see supplementary table 2).

Results

Literature search results

The initial and updated searches combined identified 2276 records from the original databasesearches. After removing duplicates, initial screening of abstracts and titles, and 67 full texts were further assessed for inclusion. At this stage, 52 full texts were excluded, leaving 15 papers for inclusion in the review. [13-27]

Out of 15 studies 5 were conducted in Australia, 6 in the UK, 2 in the USA, 1 in Canada and1 in the Netherlands. Nine collected data through interviews only, two used focus groups only, 2 used a combination of focus groups and individual interviews. One of the studies thatused interviews, had intended to use focus groups but changed data collection methodology due to difficulty recruiting to focus groups. In total, the included participants across all studieswere 366. There was marked variation between the level of information presented about participants. For studies where the information was available, 69% of participants were female and the majority were either spouses/partners or adult children of the person with dementia. Only three studies (Griffiths, Livingston and Thompson) provided data on ethnicityor deprivation. The lack of discussion of diversity in the other studies may indicate that this had either not been sufficiently considered or achieved.

Quality of reporting

CASP ratings ranged from 6 to 10 out of 10, indicating general moderate to high levels of quality in reporting qualitative research. The median of the ratings was 9, with most studiesomitting to report reflection on the relationship between the researcher and the participant.

Synthesis

Support needs of carers making proxy healthcare decisions for PwD fell into 3 main domains;informational support, practical support and emotional support. Third order constructs within this synthesis are illustrated using example first order constructs from the original papers.

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Domain 1:

Informational support needs

Carers needed a number of pieces of information to be able to make proxy decisions. This domain captures support needs regarding information on dementia itself as well as details ofthe healthcare decision being faced.

Awareness of dementia and disease trajectory

In order to be able to make confident and informed decisions for their loved ones, carers require a sound understanding of dementia, especially with regards to the disease trajectory. An absence of this understanding was a recurring theme in many of the included studies. Thereality of inevitable cognitive decline and loss of capacity only became apparent to many participants as they lived through it.

"We didn't realise what dementia meant, the implications. . . I think that people who are carersshould receive some training . . . told what to expect and what to do, before it happens, not
when it happens." -Livingston

Information about the individual decision to be madeInformation content

Shared decision making requires individuals to have sufficient information about the optionsavailable and the evidence, risks, benefits for these. In reality, carers described making decisions with incomplete information.

"They didn't say it like it were…we thought there would be one or two [side effects], but Ididn't think they'd be as severe as they were"- McWilliams

"It's also hard to know at each step, is it happening because of the disease or is she medicated? I don't know" -Debellis

Information Delivery

Carers were usually keen to involve the person with dementia as much as possible in decision making. In order to achieve that, information needed to be delivered at a slower

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thannormal pace and carers often went to efforts to translate the information into a more understandable form.

"It's really fast for me…all I could think was well if I wasn't there and you [patient] wouldn't have remembered…for somebody with dementia it needs to be slowed down" -McWilliams

In reality, the person with dementia was not usually able to make healthcare decisions due totheir complex nature, but the intention to involve them remained important.

"We start off as a joint thing to talk about what needs deciding but normally [it] ends up withme making the final [decision]."-Sinclair

Signposting

Carers often sought informational support from sources other than the health care professional. Other carers with experience of making similar decisions were particularlyuseful.

"I've got an online specialist frontotemporal dementia support forum which is an international forum which consists entirely of other carers and other patients and it's a very knowledgeableforum very well organised very well moderated. And I wish I'd found it 2 years
earlier"-Shanley

Appropriate signposting to these services can help carers to feel supported with healthcaredecisions.

Practical Support Needs

This domain focuses on the practicalities surrounding healthcare decisions. It deals with howservices can be organised to provide better support for carers. It also considers legal supportstructures.

Continuity of Care

Having a trusted health or social care professional who understood both the patient and thehealthcare system was hugely valued by carers.

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‘‘We have a lot of faith in our case manager. He knows the entire dementia network andwhatever question or needs we have, he takes care of it.’’ -Wolfs

Many carers had not experienced continuity of care but recognized how valuable it could havebeen when faced with healthcare decisions.

‘when he was initially diagnosed, it would have been great to have somebody from that pointonwards that you could have dealt with like one single person or one single number that you

could call and say, “This is happening now. What typically happens here? You know, what,what is a good decision to make here?” … So yeah it just would have been good to, to, I
suppose just to have some sort of advocacy [Right] for the families just so that you can makeinformed decisions.’-Shanley

Person Centred Care

Carers become experts in the person with dementia as an individual. They act as advocatesand their input into healthcare decisions is essential to achieve true person centred care.

Carers’ expertise in their loved one’s condition and knowledge of their preferences was routinely overlooked or dismissed by healthcare staff. This left carers feeling disempoweredand frustrated.

“there was never any communication with me, when I did try to say something they justignored me and listened to him”-Jamieson

Early planning and legal context

Proxy healthcare decisions are easier to make when carers have a clear knowledge of the person with dementia’s wishes. Encouraging families to have these discussions early in the disease trajectory whilst the person with dementia has capacity to do so facilitates future

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healthcare decision making.

"Yes, it was very, very difficult. And in hindsight, it would have been a wonderful thing to havehad them [conversations] but we didn't."*-Shanley*

These conversations are difficult and some individuals and families need a lot of support withthem. Explanation of the potential benefits for carers when they are faced with healthcare decisions in the future may help some individuals to address this.

"we tried to talk to mummy about what care she would like further down the line, but she never discuss it... she just wouldn't get into that conversation...and I suppose just when we had to start making those decisions mummy was beyond having any input really in it" -Carter

The studies were conducted in a range of countries. Each of the included countries have legalprovisions which allow individuals to formally nominate one or more individuals to act on their behalf should they lack capacity in future. In the UK, this is known as Lasting Power of Attorney (LPA). The studies in this review showed that some participants were aware of the existence of these legal frameworks. A minority had put them in place. Commonly, carers andpeople with dementia were resistant to making LPAs.

...I didn't say not at all....I wouldn't do it until much later on I think...I see it as something youhand over when its necessary, not before…-Dening

The mistaken belief that there would be opportunity to set up LPAs once dementia was moreadvanced led to many reaching a situation where the opportunity was lost.

The only thing that could happen now is Court of Protection . . . because my wife can't sign."Carer-Livingston

One technique that some families used to encourage the person with dementia to make anLPA was to make them for other family members at the same time. This allowed it to be framed as a generic safeguard in case of future eventualities rather than an individual decision necessitated by their inevitable loss of capacity.

"if we did [lasting power of attorney] for both of them [parents], it wouldn't feel like it was justfor my mum because she's dementing."-Livingston

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Emotional Support

This domain focuses on the emotional impact of making healthcare decisions by proxy andhow carers can be supported with this.

Guilt

Making proxy healthcare decisions can be an emotionally fraught process. Health care professionals need to be mindful of the impact that decision making can have on carers.Checking in with carers about the emotional burden and where necessary signposting toservices is important.

"But for a time there, I did feel like I was in charge of ending my mother's life. And I think, if Ihadn't had the psychologist to talk that through, I would have found that really difficult, reallydifficult "-Shanley

The burden of guilt was particularly difficult to manage when carers had to make decisions alone or in conflict with family members.

"My brother and sister . . . wanted the drip, antibiotics and the oxygen reinstated . . . and the doctor said 'it will flood her heart' and she died a horrific death . . . but my brother always said,'oh where there's hope' . . . There was no hope." -Livingston

Where decisions were shared, it significantly alleviated the guilt.

"My case manager explains what options there are and what they mean, and I make the decision together with my mother and my sisters. If you work as a team, you make the bestchoices.''-Wolfs

Empathic Guidance

When facing proxy decisions, carers appreciated guidance from health care professionals. Itwas important however that those decisions were not made unilaterally by professionals.

Recognition and consideration of the needs and concerns of the person with dementia andthe carer were key. This approach is described variously in the included papers, but

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one paper (Shanley) uses the elegant term empathic guidance.

"And, and I think that was, it was very, it was really helpful to have the medical staff kind of putit in, you know, they didn't beat around the bush … I really felt like they were very supportive and they were looking out for her, and they were looking out for me as well. And so I kind of, Idid put a lot of trust in what they said … And they were actually considering her needs at the time and, and so they helped me make that decision."- Shanley

Where health care professionals did not act as empathic guides and failed to demonstrate consideration of the needs of all parties, attempts to engage carers in decision making wereoften superficial and unsatisfactory.

"The hospital was no help, and I got aggravated with all of them and finally I had to call ameeting between all the doctors because I could not get anyone to give me any answers."-Truglio Londrigon

Discussion

Summary of main findings

In order to make healthcare decisions for people with dementia, carers require an understanding of the dementia itself and the likely disease trajectory. They also need information about the decision to be made, presented in a form that can be understood andthat can be shared with the person with dementia wherever possible.

Carers may be facilitated in decision making by access to support groups, especially withcarers who have faced similar decisions.

Preparation for future healthcare decision making should be facilitated early in the dementiatrajectory so that carers feel more confident about the person with dementia's wishes and sothat decisions truly are made in their best interests. Health care professionals should encourage families to formalise arrangements for future decision making using the relevant legal structures such as LPAs.

Professionals who recognize the emotional burden of decision making and act as empathicguides for carers making proxy healthcare decisions are highly valued.

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Strengths and limitations of the study

This systematic review is novel in that it explores proxy healthcare decision making at earlier stages of the disease trajectory for people with dementia, rather than focusing on end of life care or care home placement decision making. We applied rigorous quality assurance to eachstudy and identified strengths and gaps in the evidence base as well as synthesizing key themes of relevance to practice and policy. We have been guided by the report from the eMERGe reporting guidance which says there are ‘ no bespoke reporting guidance exists for meta-ethnography...Existing generic guidance for reporting qualitative evidence synthesis pays insufficient attention to reporting the complex synthesis processes of meta-ethnography’.[28] With this in mind, we have used the PRISMA checklist for both abstractand article paper in ensuring there was good practice and rigorous, comprehensive analysis,interpretation, and synthesis of the evidence.

A systematic review is by its nature dependent on the quality of the original studies. Of the 15studies in this review, only one (Livingston) provided detailed demographic data on participants. This study recruited a diverse population in terms of ethnicity, educational background, age and gender. The lack of demographic information in the other studies maysuggest a lack of diversity, but certainly suggests a lack of consideration of diversity by the authors. The studies were all conducted in English and excluded the voices of people for whom English is not their first language.

The goal of this review was to focus on the support needs of carers making proxy healthcare decisions for people with dementia. Few of the included papers focused entirely on healthcaredecisions from a carer perspective; many dealt with a range of decision types, others includedviews of other parties such as the person with dementia or the health care professional. As a result, there were often only small sections of the included papers, which yielded relevant quotes.

The thematic synthesis was conducted by 4 academic GPs; VH, CM, CH and SL. Whilst every effort was made to maintain a reflexive approach there will undoubtedly have been shared assumptions and blind spots which could have been mitigated with a more diverseresearch team.

Comparison with existing literature

Information and knowledge about

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dementia

A lack of knowledge about dementia and the disease trajectory was a common theme throughout this review. A recent systematic review of the public's knowledge and understanding of Alzheimer's disease and dementia found that there was a common misconception that dementia was a normal part of aging. [29] Carers described a need for a post-diagnosis training programme to teach them what to expect. Key within this was the importance of early advanced care planning. Many participants missed the opportunity to have advanced care planning discussions with their loved ones, in part because they did notrealise the inevitable downward trajectory of dementia with resultant loss of capacity. Proxy healthcare decisions were much easier to make and accept when carers had a clear understanding of their relative's wishes. Supporting carers and people with dementia to havethose conversations early facilitates proxy decision making later on.

Shared decision making

Shared decision making is a key component of the NHS long term plan.[4] It is widely acceptedthat shared decision making facilitates patients to make decisions that are best for them. A 2011 report by The King's Fund [5] states at the heart of decision making 'is the recognition thatclinicians and patients bring different but equally important forms of expertise to the
decision-making process. In the context of proxy-decision making, the carer should take therole of the second expert, however our review suggests that their expertise is commonly disregarded.

These findings echo the Care Quality Commission's 2016 review of how people are involved in their care, which found limited understanding, recording and monitoring of people's wishesand preferences, inadequate family and carer involvement and insufficient information and explanation of care and support options.[30] Some of the barriers to the implementation of shared decision making identified in the literature include practical factors such as a lack of continuity of care and time constraints. Additionally the challenges involved in building good rapport between both parties has been identified.[31] Some professional groups have proposedsolutions such as templates that can be used at various levels of service delivery,[32] while other authors argue that more research is still needed in the area of shared decision making in primary care in

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order to better conceptualise the barriers and solutions.[33]

Continuity of Care

Many studies have shown the benefits of continuity of care both in terms of patient experienceand clinical outcomes. [34-37] The 2018 NICE dementia guidelines recommend that people living with dementia be provided with a single named health or social care professional who isresponsible for coordinating their care and that a program of post diagnostic support be offered to carers. [38] In practice, delivery of this support is inconsistent across the UK, with multiple sectors delivering a range of services with the potential for duplication or gaps.[39] Our review found similarly inconsistent access to care coordination, but where it was available, carers valued it.

Digital Signposting

The emotional burden of decision making for a person with dementia can be enormous. Oneof the many consequences of the covid 19 pandemic is the loss of in person support and support networks. [40] Increasing use of on-line support forums offers an alternative for some, but 29% of people aged over 65 in the UK have never used the internet. [41] The movement ofservices online which has been hugely accelerated by the pandemic risks leaving those without internet access or skills unable to access a wealth of informational, practical and emotional support options.

Implications for future research and clinical practice

The NHS aspires towards making person centred care, including shared decision making, part of routine practice. Research is needed to establish how we can achieve this meaningfully for proxy decision makers, so that both they and the person with dementia haveaccess to the benefits of person centred care. Patients with dementia suffer from inevitable decline in cognition and most will reach a point where they are unable to make healthcare decisions independently. If people with dementia are to benefit from the drive towards patientcentred care, it is vital that we consider the support needs of their carers.

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Continuity of care can be difficult to achieve in the modern health system, but has undoubtedbenefits for patients with complex comorbidity.[42] Commissioners should consider how continuity can best be achieved. It should be clear to practitioners and service users who thatsingle point of contact is and how to access necessary support.

In an increasingly digital world, there is potential for widening health inequalities by reducingaccess to services for those who do not have internet access or computer literacy. A significant minority of older adults do not use the internet and policy makers and commissioners must take measures to ensure that services remain accessible to all, whilst also providing opportunities for people to improve their digital literacy.

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Data Statement No additional data are available

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Table 1: PICO

Sample	People of any age and gender who have a close relative with dementia (regardless of where they reside)
Phenomenon of interest	Proxy-decision making about a healthcare treatment
Design	Any qualitative primary, empirical research
Evaluation (Outcomes)	Experiences, support needs, barriers, facilitators of family/informal carers making proxy decisions about healthcare treatment for a relative with dementia

Table 2

Construct	Explanation	Example
First order constructs	Carers accounts of their experiences	*'it's a lot of negotiation… sometimes…it's a case of having a familiar face'* *I think, you know, when he was initially diagnosed, it would have been great to have somebody from that point onwards that you could have dealt with like one single person or one single number that you could call and say, "This is happening now. What typically happens here? You know, what, what is a good decision to make here?" … So yeah it just would have been good to, to, I suppose just to have some sort of advocacy [Right] for the families just so that you can make informed decisions.*

Second order constructs	Author's views and interpretations expressed as themes in the primary papers	Similarly, healthcare professionals unfamiliar with the family and the resident's individual wishes were also noted to cause unnecessary anxiety, again resulting in reluctance of further contact. The experience of an uncoordinated discharge process was for family and friends compounded by their own exhaustion after their hospital vigil. The tiredness was further complicated by delays in restarting community services that were accessed previously This health professional was often their GP. Having this consistent, ongoing support helped the surrogate decision- maker become aware of issues and find their way through the complex maze of service provision. It meant that surrogate decision- makers did not have to keep repeating their story and building trust with new professionals.
Third order constructs	The views and interpretations of the synthesis team	Continuity of Care Having a trusted health or social care professional who understood both the patient and the healthcare system was hugely valued by carers.

Supplementary Table 1: Data items

Author, year and country	**Stated aim**	**Participant characteristics**	**Method of data collection**	**Analytic approach**	**Decisions examined Themes**	**Setting**	**Range of capacity of person living with dementia**
Bloomer et al 2016 Australia[13]	Explore the experiences of family carers of people with dementia who are hospitalised	20 carers age range 34-92 equal gender distribution	Individual face to face interviews	Thematic analysis	Experience of transitioning through the hospital system	Hospital	Not mentioned
		mixture of spouses, adult children and friends (no details provided)					

Carter 2018 UK[14]	Exploring the experiences and preparedness of family carers for best interest decision-making of a relative living with advanced dementia	20 carers 75% female 4 siblings, 2 spouse, 13 children, 1 extended family member	Face to face interviews	Thematic analysis	Experiences and preparedness of family carers for best interest decision-making on healthcare needs	Nursing home	Not mentioned

DeBellis 2017 Australia[15]	To explore relatives' experience, knowledge and perceptions of challenging behavioural and psychological symptoms of dementia and association with antipsychotic use for persons with dementia in residential aged care.	6 participants, all F age 45-62, 4 daughters, one wife, one daughter in law	Face to face interviews	Thematic analysis	Experiences, knowledge and perceptions of antipsychotic medication use for the person with dementia	Residential care home	Not mentioned
Dening 2017 UK[16]	To understand the lived context of healthcare decision making in the past, present and future for people with dementia and their family carer	6 carers age 49-85 5 spouses, 1 adult child	Face to face interviews	Thematic analysis	Healthcare decision-making in the past, present and future	Memory clinic service	Had capacity

Elliot et al 2007 USA[17]	Investigating the decision making of family members of institutionalised older adults with advanced cognitive impairment, with particular focus on spirituality and meaning	39 family members average age 62 4 same generation (spouse/sibling) 35 adult children	Focus groups	Constant comparative method	Healthcare decisions	Nursing home	Not mentioned
Griffiths et al UK[18]	Decision-making in cancer care for people living with dementia	22 family relatives Female (64%) Relationship to participant Child 55% Spouse 32% Sibling 9% Grandchild 5%	Face to face interview with relatives	Thematic analysis	Decisions on healthcare needs	Hospital setting	Full capacity

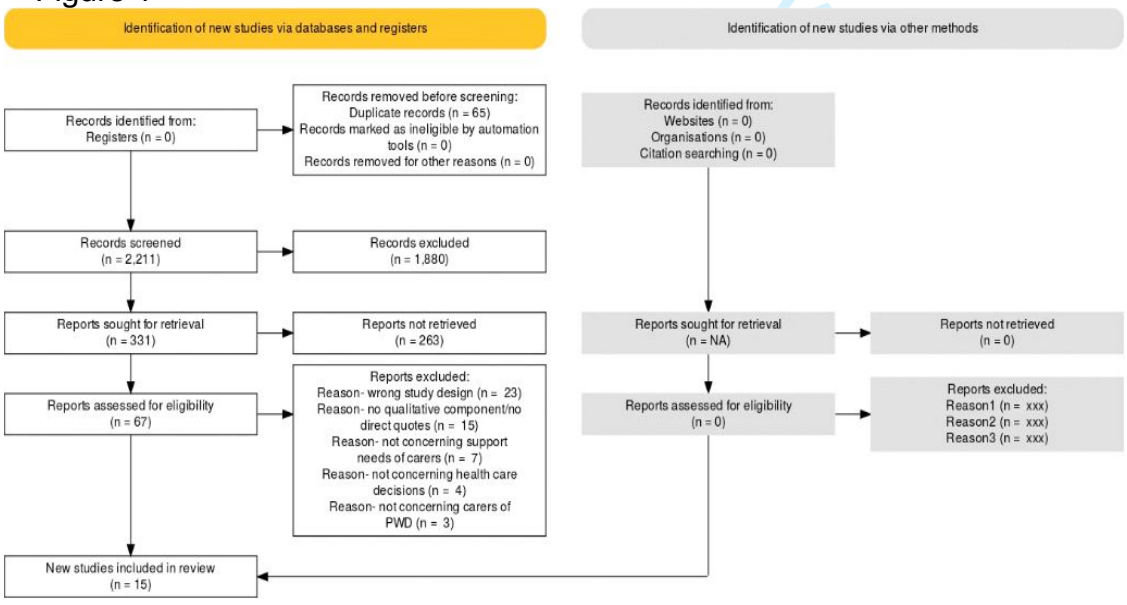
Jamieson 2016 Australia[19]	Investigating the experience of people with dementia and their carers when transitioning from home to hospital	30 carers 87% female	Started with focus groups, but as these were difficult for carers to access, changed to individual telephone interviews	Thematic analysis	Decisions on health and social care needs during transitions home from hospital	Non-residential	Not mentioned
Livingston 2010 UK[20]	Exploring how carers make decisions for people with dementia who lack capacity	Focus groups: 43 participants Individual interviews: 46	Focus groups to generate list of difficult decisions, Individual interviews to explore in depth	Grounded theory	Decisions on health and social needs	Primary care clinics	Not mentioned

McWilliams 2018 UK[21]	To investigate cancer-related information needs and treatment decision making experiences of people with cancer-dementia, their informal caregives and oncology HCPs	9 carers mean age 73.6 44% female 56% spouse, 44% child	Face to face interviews	Thematic analysis	Decisions on healthcare needs	Hospital setting	PwDranged from full capacity to no capacity

Shanley et al 2017 Australia[22]	Providing support to surrogate decision-makers for people living with dementia: Healthcare professional, organisational and community responsibilities	34 surrogate decision makers. 79 % female 85% adult children mean age 57.4	Individual face to face or telephone interviews	Thematic analysis	Decisions included residential care, giving up driving, consenting to medical treatment and end of life	Community or in a residential aged care setting	Not mentioned
Sinclair et al 2018 Australia[23]	Explore how couples with dementia experience healthcare, lifestyle, and everyday decision-making	15 carers Aged 59-82 60% female All spouse partners	Face to face interviews, 6 individual, 9 couple dyad interviews	Thematic analysis	Decisions on healthcare, lifestyle, and everyday decision-making	Community and residential care settings	Not mentioned
Thompson et al 2021 Canada[24]	Clarifying the information and support needs of family caregivers of nursing home residents with advancing dementia	17 family caregivers Female (70.6%) Male (29.4%)	Face to face interviews with bereaved family caregivers of persons with dementia,	Thematic analysis	Decisions on healthcare needs	Nursing home setting	Not mentioned

Truglio-Londrigan 2018 USA[25]	To identify Alzheimers Disease specific decisions that past and present caregivers faced over the disease trajectory	13 carers 85% female 9 adult children, 3 spouses	Focus groups	Thematic analysis	Decisions on health and social care needs	Community setting	Not mentioned
Walker 2001 UK[26]	Investigating how we facilitate carer's involvement in decision making?	20 carers of people with dementia No further details on participants provided	Face to face individual interviews	Constant comparative method	Decisions on healthcare needs	Hospital setting	Not mentioned
Wolfs 2012 The Netherlands[27]	Gaining carer's insights into the decision making process about treatment and care in dementia patients	26 carers 50% female 11 spouses, 12 adult children,3 other	Brief individual interviews inform interview schedule, then focus groups	Grounded theory	Decisions on healthcare needs	Community setting	Not mentioned

Figure 1



Supplementary Table 2: CASP ratings

	Bloomer	Carter	DeBellis	Deming	Elliott	Griffiths	Jamieson	Livingston	McWilliams	Shanley	Sinclair	Thompson	Truglio-Londrigon	Walker	Wolfs
Was there a clear statementof aims ofthe research?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Is qualitative methodology appropriate?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the research design appropriate to address the aims of the research?	yes	yes	no	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the recruitme nt strategy appropriate to the aims of the research	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data collected in a way that addressedthe research issue?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Has the relationship between researcher and participant s been adequately considere d?	no	no	no	No	no	yes	no	no	no	no	no	yes	no	no	yes
Have ethical issues been taken into considerat ion?	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data analysis sufficiently rigorous?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes

Is there a clear statementof findings?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
How valuable is the research?	high	high	mediu m	medi u m	low	High	high	high	high	high	high	High	high	mediu m	high

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30th July 202

The Editor-in-chief,

BMJ Open,

Dear

Thank you for your email, in which you suggested that our manuscript might be accepted for publication in the bmjopen, subject to some revisions. We have taken into account your formatting requirements and the sets of reviewer comments and have revised our manuscript accordingly.

Please find below our point-by-point response to each of the reviewers comments quoting the text and line numbers where appropriate. As requested, we have also uploaded a revised marked copy with highlighted text where alterations have been made. Thank you for the supportive feedback and peer review process which we believe has strengthened the paper.

We are very pleased to re-submit this manuscript.

Kind regards,

Victoria Hodges

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Comment from editor	
Please revise the formatting of your abstract so that it includes the following sections: Objectives >> Design >> Data Sources >> Eligibility Criteria >> Data extraction and synthesis >> Results >> Conclusions. Please see the following published abstract as an example: http://bmjopen.bmj.com/content/8/3/e019438.	Thank you very much for this suggestion and the paper is now more compliant to the bmj open format. This format has been re-written to reflect this format in the abstract.
-Please add bullet points to your 'Strengths and limitations' section.	This has been edited and bullet points now added to the 'Strengths and limitations of study' sub-heading, page 3.
-Please update your search results to the present time.	Thank you and we have edited the search results updated on 29th June 2021. 2 additional titles added to the review as a result.
-Please include, as a supplementary file, the precise search strategy (or strategies) for one database, including any limits and filters.	Clarification of search strategy employed added to the search strategy sub-section, page 5.
-Along with your revised manuscript, please include a copy of the full PRISMA checklist (rather than one adapted for kinesiology) indicating the page/line numbers of your manuscript where the relevant information can be found (http://www.prisma-statement.org/).	Copy of the PRISMA checklist as appendix 1 attached and this indicates the page of our manuscript where the relevant information can be found
Comments from reviewer 1 and 2	
Comments from reviewer 1	
Please use PRISMA to write your abstract add eligibility criteria in the abstract ;* Put the Background subsection before objective	Thank you for this suggestion and we have now edited to use PRISMA and to put the background sub-section before objective
Briefly present the eligibility criteria in the Methods subsection of the abstract	Thank you and this section has now been edited and updated under the sub-heading 'Eligibility criteria' on page 1.

Commented [1]: usually you ned a version of the document which shows the markups done or direct comments which link to this table

Specify the date of the documentary research	We have now edited this and the date updated: Searches were repeated on the 22nd June 2021.' This updated search identified 2 extra papers included now in the table and analysis of this additional data reinforced our findings and did not change the key findings of the review.
No "study selection" subsection	This has now been included as a sub-section 'Study selection on page 6.
Delete the conclusion subsection	Sub-section deleted from page 23.
Add a discussion subsection	We have now added a sub-section 'Discussion' added on page 20
Objective is not clear, would need to be rephrased The objective does not clearly mention the topic of dementia, unlike the objective stated in the abstract section. In addition, the authors generalize the scope of the review to care decisions other than those relating to end-of-life care "This systematic review and meta-ethnography of qualitative studies aimed to review the existing literature and develop new understanding of the support needs of carers facing health care decisions other than those relating to end of life care and to inform practitioners who may be in a position to provide support. "	The study aim has been added to the section Search strategy and selection strategy on page 5*'The objective of this study was to review the existing literature on the support needs of carers acting as proxy healthcare decision makers for people with dementia (PwD).'* In our exclusion criteria we have made it clear that this systematic review excludes papers where the sole purpose of the study was End of Life Care on page 5.
Methods 7-9 * Structure the "Method" section according to the PRISMA guideline	Thank you for highlighting this issue. We have added an additional sentence to the strengths and limitations on page 14 of our research and hope this is helpful in clarifying our approach and structure of results section: We have been guided by the report from the eMERGe reporting guidance which says there are *' no bespoke reporting guidance exists for meta-ethnography...Existing generic guidance for reporting qualitative evidence synthesis pays insufficient attention to reporting the complex synthesis processes of meta-ethnography.'* With

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	this in mind we have not used the PRISMA in structuring the headings of the paper but have used the PRISMA checklist for both abstract and article paper in ensuring there was good practice and rigorous, comprehensive analysis, interpretation, and synthesis of the evidence.' (reference: France, E.F., Cunningham, M., Ring, N. et al. Improving reporting of meta-ethnography: the eMERGe reporting guidance. BMC Med Res Methodol 19, 25 (2019). https://doi.org/10.1186/s12874-018-0600-0)..
Clearly specify the eligibility criteria for the studies included in the review	This has now been specified under the sub-heading 'Eligibility criteria' on page 5. Eligibility criteria Inclusion criteria: a) People of any age and gender who have a close relative with dementia (regardless of where they reside), b) Proxy-decision making about a healthcare treatment c) Any primary, empirical research Qualitative, mixed methods Exclusion criteria: a) Published prior to 1995 b) Non-English language c) Only report proxy decisions not related to healthcare d) End of Life e) Reviews, editorials, dissertations, meeting abstracts and correspondence
The research strategy is to focus on dementia or loss of cognitive ability, which does not seem to correspond exactly to the objective as formulated in the "Introduction" section.	Thank you for this comment. This has been edited to reflect the focus on dementia in the rationale and research objectives on page 4. The term 'loss of cognitive ability has now been deleted.
I suggest clarifying the elements of the PICO and reformulating the objective precisely according to the PICO	Thank you and we have now created a PICO table sowing the objectives. Table 1.
Describe the study selection process before data extraction	Edited and a new section 'search strategy and selection criteria' on page 5.
Results 9-15 * Update the PRISMA flow	Updated and added to the

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chart according to the updated version PRISMA 2020 http://www.prisma-statement.org/PRISMAStatement/FlowDiagram	supplementary PRISMA file.
Rename the sub-sections of the result part according to the PRISMA2020 guideline	Thank you for highlighting this issue. We have added an additional sentence to the strengths and limitations on page 14 of our research and hope this is helpful in clarifying our approach and structure of results section: Please see above re addition to the strengths and limitations of the research. We have been guided by the report from the eMERGe reporting guidance which says there are *' no bespoke reporting guidance exists for meta-ethnography...Existing generic guidance for reporting qualitative evidence synthesis pays insufficient attention to reporting the complex synthesis processes of meta-ethnography'* (reference: France, E.F., Cunningham, M., Ring, N. et al. Improving reporting of meta-ethnography: the eMERGe reporting guidance. BMC Med Res Methodol 19, 25 (2019). https://doi.org/10.1186/s12874-018-0600-0). With this in mind we have not used the PRISMA in structuring the headings of the results section paper but have used the PRISMA checklist for both abstract and article paper in ensuring there was good practice and rigorous, comprehensive analysis, interpretation, and synthesis of the evidence.
In view of the results presented, this is a systematic review presenting a synthesis of qualitative data based on the 7 steps of Noblit and Hare for performing a meta-ethnography. We therefore suggest simplifying the title of the article.	We are grateful for this comment as we have now edited the title, which gives it more focus and clarity. Title now edited 'Support Needs of Carers Making Proxy Healthcare Decisions for People with dementia; a systematic review based on the Noblit and Hare meta-ethnographic synthesis of qualitative studies.'

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Discussion 15-19 * Reorganize the discussion section according to PRISMA 2020	Thank you for highlighting this issue. We have added an additional sentence to the strengths and limitations on page 14 of our research and hope this is helpful in clarifying our approach and structure of results section: We have been guided by the report from the eMERGe reporting guidance which says there are ' no bespoke reporting guidance exists for meta-ethnography...Existing generic guidance for reporting qualitative evidence synthesis pays insufficient attention to reporting the complex synthesis processes of meta-ethnography' (reference: France, E.F., Cunningham, M., Ring, N. et al. Improving reporting of meta-ethnography: the eMERGe reporting guidance. BMC Med Res Methodol 19, 25 (2019). https://doi.org/10.1186/s12874-018-0600-0). With this in mind we have not used the PRISMA in structuring the headings of the paper but have used the PRISMA checklist for both abstract and article paper in ensuring there was good practice and rigorous, comprehensive analysis, interpretation, and synthesis of the evidence. We followed the journal format in author guidelines for the discussion section.
Rename the last paragraph "Conclusion" instead of "In conclusion"	Thanks for the suggestion, however, the conclusion section has been deleted now and added to discussion section as Implications for future research and clinical practice section on page 16, as this was suggested by reviewer 1
Appendices * Structure the flow chart according to the PRISMA template http://www.prisma-statement.org/PRISMAStatement/FlowDiagram	New template now added
The conclusion as formulated does not mention the main needs identified by the study	The main needs have now been incorporated into the discussion section and implications for future research and clinical practice. Conclusion section deleted now and added to discussion

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	section and Implications for future research and clinical practice section, page 16
Reviewer 2	
Provide the review protocol questions in the paper even though they are in the protocol.	Thank you and we have addressed this comment by adding a sub-section on research questions in the introduction section, page 4
Provide more information about excluded papers - it is normal to provide numbers against categories such as - not the right patient/population	Thank you and the details of this are now in the PRISMA diagram
3. a) Review the analysis section to provide more clarity about the process - this section could benefit from referencing and examples to show how the coding through the 3 order constructs worked. b) Provide more summary details about the included papers. This information will help the reader to navigate the results. I would expect to have seen a summary of the range of decisions the paper covered, the settings of the decisions (acute, community, primary health), the range of capacities of the people with dementia.	Thank you, Table 2 with an example of how the coding for the 3 order constructs was applied in this study and has now been included to page 8. Table 3 has been edited to show the range of decisions, the setting and capacity of the dementia patients.
5. I also feel a slightly more academically critical lens was needed regarding the evidence for shared decision and stronger selection of articles for referencing.	Thank you and we have taken your advice, appraised and added additional references in our discussion on page 15 and we think the argument is stronger now. Additional references: 31. Elissa M. Abrams, Marcus Shaker, John Oppenheimer, Ray S. Davis, Don A. Bukstein, Matthew Greenhawt, The Challenges and Opportunities for

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	Shared Decision Making Highlighted by COVID-19, The Journal of Allergy and Clinical Immunology: In Practice,2020; 8(8)2474-2480 32. Légaré F, Canada Research Chair in Implementation of Shared Decision Making in Primary Care. Interprofessional approaches to shared decision making in primary care: Advancing theories, frameworks, methods and measurements. 33. Murray E, Charles C, Gafni A. Shared decision-making in primary care: tailoring the Charles et al. model to fit the context of general practice. Patient Educ Couns. 2006 Aug;62(2):205-11. doi: 10.1016/j.pec.2005.07.003. Epub 2005 Aug 31. PMID: 16139467.

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title – page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Completed and added as supplementary documents
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction. Last paragraph. Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	In abstract page 1 and search strategy section page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Eligibility criteria page 5
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Information sources page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Search strategy page 5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Study selection and quality appraisal page 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Quality appraisal page 6 and data extraction page 6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Data extraction page 6 and Table 2 & 4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Quality of reporting page 8 and CASP rating table (Table 2)
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Data extraction page 6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	PICO table 1
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	CASP rating table (table 4)
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Data extraction page 6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Table 3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Data extraction page 6

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PRISMA 2020 Checklist

	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Data extraction page 6
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Bias assessment page 7, Data extraction page 6 Strengths and limitations page 2
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Synthesis page 8
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Data extraction page 6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	PRISMA diagram figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Strengths and limitations page 13
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Synthesis page 8
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Strength and limitations page 13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Summary of findings page 13 and comparison with existing literature page 14
	23b	Discuss any limitations of the evidence included in the review.	Strengths and limitations page 13
	23c	Discuss any limitations of the review processes used.	
	23d	Discuss implications of the results for practice, policy, and future research.	Implications for future research and clinical practice page 16
OTHER INFORMATION			
Registration and	24a	Provide registration information for the review, including register name and registration number, or state that the	registration number CRD42020124485

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protocol		review was not registered.	
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	https://www.crd.york.ac.uk/PROSPERO
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 2
Competing interests	26	Declare any competing interests of review authors.	Page 2
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	CASP Table 4

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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BMJ Open

Support Needs of Carers Making Proxy Healthcare Decisions for People with dementia; a systematic review based on the Noblit and Hare meta-ethnographic synthesis of qualitative studies.

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Primary Subject Heading:	Geriatric medicine
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Keywords:	Dementia < NEUROLOGY, GERIATRIC MEDICINE, Health policy < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Quality in health care < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Old age psychiatry < PSYCHIATRY

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Support Needs of Carers Making Proxy Healthcare Decisions for People with dementia; a systematic review based on the Noblit and Hare meta-ethnographicsynthesis of qualitative studies.

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Key Words: Dementia; Caregivers; Decision making, shared; Advanced care planning

Word count: 4583

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Abstract

Background There are 50 million dementia sufferers worldwide. Decisions about healthcareoften need to be made when the person with dementia lacks capacity to do so.

Understanding the support needs of carers acting as proxy healthcare decision makers will be vital in improving the decision making process for people with dementia and addressingthe holistic needs of carers.

Objective The objective of this study was to review the existing literature on the supportneeds of carers acting as proxy healthcare decision makers for people with dementia.

Design A qualitative systematic review protocol was published on PROSPERO. The CriticalAppraisal Skills Programme (CASP) checklist was used to appraise study quality. A meta-ethnographic synthesis was performed to develop third order constructs.

Data Sources A search was conducted using three online databases (MEDLINE, CINAHLand PsycINFO).

Eligibility criteria Fifteen studies met the inclusion criteria: Primary qualitative research involving carers of people with dementia who had been involved in making proxy healthcaredecisions.

Data extraction and synthesis Two independent researchers conducted validity assessments for each paper selected for inclusion and discrepancies were resolved by discussions with a third reviewer. Nvivo software was used and conceptual findings fromstudy papers lead to interpretations of findings by the team.

Results From the 15 papers included in the study three main domains arose from the meta-ethnography; informational, practical and emotional. Informational support needs included information about dementia itself and the anticipated disease trajectory. Practicalneeds included continuity of care, person centred care and the use of legal frameworks.

Emotional support included recognising the guilt that healthcare decisions can provoke andthe importance of providing guidance in an empathic manner**.**

Conclusions This meta-ethnography highlights opportunities for health care professionals and policy makers to improve experiences of carers making proxy

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healthcare decisions forpeople with dementia.

Strengths and Limitations of this study

- To our knowledge this is the first systematic review to focus on the experience of proxy healthcare decision making for carers of people with dementia.
- By translating results of multiple studies, we make the results more readily availableto practitioners, policy makers and researchers.
- Although there was a comprehensive search strategy with inclusion of internationalpapers, the study did not include non-English language publications.
- Studies may also have been missed due to the variety of terminology used in this field.

Funding Statement

This work was supported by The RCGP Scientific Foundation Board, grant number SFB2018-07

Ethics Statement

Ethics approval was not required as there was no patient involvement

Competing interests

None declared.

Acknowledgements

The authors would like to thank Anthea Tucker for help with search strategies and Professor Christopher Burton for advice on the manuscript.

Introduction

Dementia affects an estimated 50 million people worldwide with 850,000 sufferers in the UK. As the UK population ages and deaths from other causes are avoided due to

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improvements inprevention and treatment, dementia prevalence is likely to rise. The number living with dementia in the UK is expected to almost double to 1.6 million in the next 10 years.[1] Dementiais the leading cause of death in the UK,[2] but the disease often runs a protracted course over many years and most people affected will have other significant health problems. Over 90% of dementia sufferers have at least one comorbidity.[1] Approximately 700,000 people in the UK provide informal care for someone with dementia. It is a particular difficult and stressful condition to care for.[3]

Decisions frequently need to be made about the healthcare of people with dementia. These range from decisions to undergo testing to commencing or continuing treatment for dementiaor for other conditions. Dementia can be worsened by the disruption to routine and changes to physical health that may result from healthcare decisions. This must be balanced against the potential gain from accepting care making these decisions complex. Carers are often required to be involved in proxy decision making about healthcare as an individual's capacityto make decisions declines.

Shared decision making is a key feature of personalized care which forms part of the NHS long term plan.[4] Shared decision making ensures that individuals are supported to make decisions that are right for them. It is a collaborative process through which a clinician supports a patient to reach a decision about their treatment. The conversation brings together: the clinician's expertise, such as treatment options, evidence, risks and benefits andwhat the patient knows best: their preferences, personal circumstances, goals, values and beliefs. Shared decision making is known to lead to more realistic expectations, a better match between individuals' values and treatment choices, and fewer unnecessary interventions.[5] The NHS long term plan states that personalized care will underpin care for people with dementia.[4] Understanding the support needs of carers acting as proxy decision makers will be vital if people with dementia who have lost decision making capacity and their carers are to benefit from the move towards shared decision making. The rationale for this systematic review and meta-ethnography of qualitative studies was to review the existing literature with a view to developing new understanding of the support needs of carers for people with dementia facing healthcare decisions and to inform practitioners who may be in a position to provide support. Most published research on makingdecisions by proxy for people with dementia has focused on end of life care [6-7] and nursing home placement. [8-10] Decision making about healthcare for comorbidities

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earlier in the disease trajectory has been less well studied.

Research Objectives

- Describe family carer experiences of proxy decision making about healthcare treatment fora relative with dementia excluding decisions about end of life care.
- Identify barriers and facilitators family carers experience when making proxy decisionsabout healthcare treatment for a relative with dementia.

Method

Patient and Public Involvement

PPI work was conducted with a dementia-specific PPI group based at the University of Sheffield. This shaped an initial scoping literature review and development of the researchquestion.

Eligibility criteria

Inclusion criteria:

a) People of any age and gender who have a close relative with dementia(regardless of where they reside)

b) Proxy-decision making about a healthcare treatment

c) Any primary, empirical research; qualitative, mixed methods

Exclusion criteria:

a) Published prior to 1995

b) Non-English language

c) Only report proxy decisions not related to healthcare

d) End of Life

e) Reviews, editorials, dissertations, meetingabstracts and correspondence

Information sources

We searched MEDLINE, PsycINFO, and CINAHL Searches were limited by date from 1st January 1995 to 1st May 2020. Searches were restricted to English language. Searches were repeated on the 22nd June 2021.

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Search strategy and selection criteria

The objective of this study was to review the existing literature on the support needs of carers acting as proxy healthcare decision makers for people with dementia. The protocol for this systematic review was published in the international prospective register of systematic reviews on 5 June 2020, registration number CRD42020124485. https://www.crd.york.ac.uk/PROSPERO

Qualitative studies that clearly reported qualitative findings were eligible for inclusion if theydiscussed proxy healthcare decision making.

Target searches using Boolean operators combined terms and Medical Subject Headings(MESH) related to

- carer or caregiver or informal carer or family carer or relative AND
- Decision making or Proxy or Decide or Surrogate decision mak* or 3rd party decisionmak* AND
- dementia or Alzheimer's AND
- Qualitative

Study selection Studies were excluded if they focused entirely on decisions about nursing orhome placement or end of life care and the steps of study selection are represented in the PRISMA diagram (figure 1).

Quality Appraisal

The qualitative version of the Critical Appraisal Skill Programme (CASP) (Critical Appraisal Skills Programme, 2018) [11] was used to assess methodological quality of the included studies by VH with a sample of papers also being assessed by SL. Discrepancies were resolved by discussion. Quality ratings informed data synthesis and were not used as exclusion criteria. The CASP was also used to reduce the risk of reporting bias and discussions of the validity ofeach study was done during data extraction and synthesis.

Data Extraction

Key features of each study were extracted by VH and independently checked by SL and

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CM.For each study, the following were extracted: author, year of publication, country of origin, stated aim, participant characteristics, method of data collection, qualitative methodology and analytical approach. The outcomes that were sought in the papers are summarized in Table 1 and these features can be found in Table 2.

Table 1: PICO

Sample	People of any age and gender who have a close relative with dementia (regardless of where they reside)
Phenomenon of interest	Proxy-decision making about a healthcare treatment
Design	Any qualitative primary, empirical research
Evaluation (Outcomes)	Experiences, support needs, barriers, facilitators of family/informal carers making proxy decisions about healthcare treatment for a relative with dementia

Table 2: Features of papers

Author, year and country	Stated aim	Participant characteristics	Method of data collection	Analytical approach	Themes/Decisions examined	Setting	Rnage of capacity of Person with dementia
Bloomer et al 2016 Australia [13]	Explore the experiences of family carers ofpeople with dementia who are hospitalised	20 carers age range34-92 equal genderdistribution mixture of spouses, adultchildren and friends (no details provided)	Individualface to face interviews	Thematicanalysis	Experience of transitioning through the hospital system	Hospital	Not mentioned
Carter 2018 UK[14]	Exploring the experiences and preparedness of family carersfor best interest decision-making of a relative living with advanced dementia	20 carers 75% female 4 siblings, 2 spouse, 13 children, 1 extended familymember	Face to face interviews	Thematicanalysis	Experiences and preparedness of family carers for best interest decision-makingon healthcare needs	Nursinghome	Not mentioned
DeBelli s 2017 Australia [15]	To explore relatives' experience, knowledge and perceptions of challenging behavioural and psychological symptoms of dementia and association with antipsychotic use for personswith dementia in residential aged care.	6 participants all F age 45-62, 4 daughters, one wife, one daughter in law	Face to face interviews	Thematicanalysis	Experiences, knowledge and perceptions of antipsychotic medication use for the person with dementia	Residential care home	Not mentioned
Dening 2017 UK[16]	To understandthe lived context of healthcare decision making in the past, present and future for people with	6 carers age 49-85 5 spouses, 1adult child	Face to face interviews	Thematicanalysis	Healthcare decision-making inthe past, present and future	Memor yclinic service	Had capacity to participate in the study

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	dementia and their family carer						
Elliot 2007 USA [17]	Investigating the decision making of family members of institutionalisedolder adults with advanced cognitive impairment, with particular focus on spirituality and meaning	39 familymembers average age 62 4 same generation (spouse/sibling) 35 adultchildren	Focus groups	Constant comparative method	Healthcare decisions	Nursing home	Not mentioned
Griffiths 2020 UK [18]	Decision-making in cancer care for peopleliving with dementia	22 familyrelatives Female (64%) Relationship toparticipant Child 55% Spouse 32% Sibling 9% Grandchild 5%	Face to face interviewwith relatives	Thematic analysis	Decisions on healthcare needs	Hospital setting	Full capacity
Jamieson 2016 Australia [19]	Investigating the experienceof people with dementia and their carers when transitioning from home to hospital	30 carers 87% female	Started with focus groups, but as these were difficult forcarers to access, changed to individual telephone interviews	Thematic analysis	Decisions on health and social care needs during transitions home from hospital	Decisions on health and social care needs during transitions home from hospital	Not mentioned
Livingston 2010 UK [20]	Exploring how carers make decisions for people with dementia wholack capacity	Focus groups:43 participants Individual interviews: 46	Focus groups to generate list of difficult decisions, Individualinterviews to explorein depth	Grounded theory	Decisions on health and socialneeds	Primary care clinics	Not mentioned
McWilliams 2018 UK [21]	To investigate cancer-related information needs and treatment decision making experiences of people with cancer-dementia, their informalcaregives and oncology HCPs	9 carers mean age 73.6 44% female 56% spouse, 44% child	Face to face interviews	Thematic analysis	Decisions on healthcare needs	Hospital setting	People with dementia ranged from having full capacity tono capacity
Shanley 2017 Australia [22]	Providing support to surrogate decision- makers for people living with dementia: Healthcare professional, organisational and community responsibilities	34 surrogatedecision makers. 79 % female 85% adultchildren mean age 57.4	Individualface to face or telephone interviews	Thematic analysis	Decisions included residential care, giving up driving, consenting to medical treatment and end of life	Community or in a residential aged care setting	Not mentioned

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Sinclair 2018 Australia [23]	Explore how couples with dementia experience healthcare, lifestyle, and everyday decision-making	15 carers Aged 59-82 60% female All spousepartners	Face to face interviews,6 individual,9 couple dyad interviews	Thematic analysis	Decisions on healthcare, lifestyle, and everyday decision-making	Community and residential care settings	Not mentioned
Thompson 2021 Canada [24]	Clarifying the information and support needs of family caregivers of nursing home residents with advancing dementia	17 family caregivers Female (70.6%) Male (29.4%)	Face to face interviewswith bereaved family caregiversof persons with dementia,	Thematic analysis	Decisions on healthcare needs	Nursing home setting	Not mentioned
Truglio-Londrigan 2018 USA [26]	To identify Alzheimers Disease specific decisions thatpast and present caregivers faced over thedisease trajectory	13 carers 85% female 9 adult children, 3 spouses	Focus groups	Thematic analysis	Decisions on health and socialcare needs	Community setting	Not mentioned
Walker 2001 UK [26]	Investigating how we facilitate carer's involvement in decision making?	20 carers of people with dementia No further details on participantsprovided	Face to face individual interviews	Constant comparative method	Decisions on healthcare needs	Hospital setting	Not mentioned
Wolfs 2012 The Netherlands [27]	Gaining carer's insights into the decision making processabout treatmentand care in dementia patients	26 carers 50% female 11 spouses, 12 adult children,3 other	Brief individual interviewsinform interview schedule, then focusgroups	Grounded theory	Decisions on healthcare needs	Community setting	Not mentioned

We followed Noblit and Hare's seven steps of meta-ethnography. [12] In a meta ethnographicanalysis, information is classified as a first, second and third order constructs. First order constructs are direct quotes within studies, second order constructs are authors' interpretations within the primary studies, third order constructs are interpretations synthesized by the reviewers from second order

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constructs.

VH and SL read each paper repeatedly to familiarise themselves before performing line by line coding of all first and second order constructs from the results and discussion sections of each paper. The research team met regularly throughout the review process to discuss the first and second order constructs. These were clustered into candidate third order constructs. The next stage of the meta-ethnography was to translate the studies into each other. This was done by focusing on each candidate third order construct in turn and compiling lists of first and second order constructs that related to it. Where candidate third order constructs were not robustly supported by the data they were rejected. A process of constant comparison between papers was conducted leading to the development of concise statements for each construct (synthesising translations). Both contradictory and directly comparable findings were considered together to develop a line of argument synthesis. An example of how this was applied in this study is seen in this table 3 below.

Table 3: Constructs

Construct	Explanation	Example
First order constructs	Carers accounts of their experiences	*'it's a lot of negotiation… sometimes…it's a case of having a familiar face'* *I think, you know, when he was initially diagnosed, it would have been great to have somebody from that point onwards that you could have dealt with like one single person or one single number that you could call and say, "This is happening now. What typically happens here?* *You know, what, what is a good decision to make here?" … So yeah it just would have been good to, to, I suppose just to have some sort of advocacy [Right] for the families just so that you can make informed decisions.*

Second order constructs	Author's views and interpretations expressed as themes in the primary papers	Similarly, healthcare professionals unfamiliar with the family and the resident's individual wishes were also noted to cause unnecessary anxiety, again resulting in reluctance of further contact. The experience of an uncoordinated discharge process was for family and friends compounded by their own exhaustion after their hospital vigil. The tiredness was further complicated by delays in restarting community services that were accessed previously This health professional was often their GP. Having this consistent, ongoing support helped the surrogate decision- maker become aware of issues and find their way through the complex maze of service provision. It meant that surrogate decision-makers did not have to keep repeating their story and building trust with new professionals.
Third order constructs	The views and interpretations of the synthesis team	Continuity of Care Having a trusted health or social care professional who understood both the patient and the healthcare system was hugely valued by carers.

Bias assessment

Two independent researchers conducted validity assessments for each paper selected for inclusion and discrepancies were resolved by discussions with a third reviewer. Nvivo software was used and conceptual findings from study papers lead to interpretations of findings by the team. A thorough CASP for qualitative studies was done to reduce for other biases (see table 4).

Table 4: CASP ratings

Study	Bloomer	Carter	DeBellis	Dening	Elliott	Griffiths	Jamieson	Livingston	McWilliams	Shanley	Sinclair	Thompson	Trugli o-Londrigon	Walker	Wolfs

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Was there a clear statement of aims of the research?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Is qualitative methodology appropriate?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the research design appropriate to address the aims of the research?	yes	yes	no	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the recruitment strategy appropriate to the aims of the research?	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data collected in a way that addressed the research issue?	yes	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Has the relationship between researcher and participants been adequately considered?	no	no	no	no	no	yes	no	no	no	no	no	no	no	no	yes

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Have ethical issues been taken into consideration?	yes	yes	yes	no	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Was the data analysis sufficiently rigorous?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
Is there a clear statement of findings?	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes	yes
How valuable is the research?	high	high	medium	medium	low	high	high	high	high	high	high	high	high	medium	high
CASP rating	9	9	8	8	6	10	9	9	9	9	9	9	9	8	10

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Results

Literature search results

The initial and updated searches combined identified 2276 records from the original databasesearches. After removing duplicates, initial screening of abstracts and titles, and 67 full texts were further assessed for inclusion. At this stage, 52 full texts were excluded, leaving 15 papers for inclusion in the review. [13-27]

Out of 15 studies 5 were conducted in Australia, 6 in the UK, 2 in the USA, 1 in Canada and1 in the Netherlands. Nine collected data through interviews only, two used focus groups only, 2 used a combination of focus groups and individual interviews. One of the studies that used interviews had intended to use focus groups but changed data

collection methodology due to difficulty recruiting to focus groups. In total, the included participants across all studies were 366. There was marked variation between the levels of information presented about participants. For studies where the information was available, 69% of participants were female and the majority were either spouses/partners or adult children of the person with dementia. Only three studies (Griffiths, Livingston and Thompson) provided data on ethnicity or deprivation. The lack of discussion of diversity in the other studies may indicate that this had either not been sufficiently considered or achieved.

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Quality of reporting

CASP ratings ranged from 6 to 10 out of 10, indicating general moderate to high levels of quality in reporting qualitative research. The median of the ratings was 9, with most studies omitting to report reflection on the relationship between the researcher and the participant.

Synthesis

An illustration of how constructs were developed is provided in table 2. Support needs of carers making proxy healthcare decisions for people with dementia fell into 3 main domains;informational support, practical support and emotional support. Third order constructs within this synthesis are illustrated using example first order constructs from the original papers.

Domain 1:

Informational support needs

Carers needed a number of pieces of information to be able to make proxy decisions. This domain captures support needs regarding information on dementia itself as well as details of the healthcare decision being faced.

Awareness of dementia and disease trajectory

In order to be able to make confident and informed decisions for their loved ones, carers require a sound understanding of dementia, especially with regards to the disease

trajectory. An absence of this understanding was a recurring theme in many of the included studies. The reality of inevitable cognitive decline and loss of capacity only became apparent to many participants as they lived through it.

"We didn't realise what dementia meant, the implications. . . I think that people who are carersshould receive some training . . . told what to expect and what to do, before it happens, not when it happens." -Livingston

Information about the individual decision to be made

Information content

Shared decision making requires individuals to have sufficient information about the optionsavailable and the evidence, risks, benefits for these. In reality, carers described making decisions with incomplete information.

"They didn't say it like it were…we thought there would be one or two [side effects], but Ididn't think they'd be as severe as they were"- McWilliams

"It's also hard to know at each step, is it happening because of the disease or is she medicated? I don't know" -Debellis

Information Delivery

Carers were usually keen to involve the person with dementia as much as possible in decision making. In order to achieve that, information needed to be delivered at a slower thannormal pace and carers often went to efforts to translate the information into a more understandable form.

"It's really fast for me…all I could think was well if I wasn't there and you [patient] wouldn't have remembered…for somebody with dementia it needs to be slowed down" -McWilliams

In reality, the person with dementia was not usually able to make healthcare decisions due to their complex nature, but the intention to involve them remained important.

"We start off as a joint thing to talk about what needs deciding but normally [it] ends up

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withme making the final [decision].”-Sinclair

Signposting

Carers often sought informational support from sources other than the health care professional. Other carers with experience of making similar decisions were particularly useful.

“I’ve got an online specialist frontotemporal dementia support forum which is an international forum which consists entirely of other carers and other patients and it’s a very knowledgeableforum very well organised very well moderated. And I wish I’d found it 2 years
earlier”-Shanley

Appropriate signposting to these services can help carers to feel supported with healthcare decisions.

Practical Support Needs

This domain focuses on the practicalities surrounding healthcare decisions. It deals with howservices can be organised to provide better support for carers. It also considers legal support structures.

Continuity of Care

Having a trusted health or social care professional who understood both the patient and thehealthcare system was hugely valued by carers.

‘‘We have a lot of faith in our case manager. He knows the entire dementia network andwhatever question or needs we have, he takes care of it.’’ -Wolfs

Many carers had not experienced continuity of care but recognized how valuable it could havebeen when faced with healthcare decisions.

‘when he was initially diagnosed, it would have been great to have somebody from that pointonwards that you could have dealt with like one single person or one single number that you could call and say, “This is happening now. What typically happens here? You know, what,what is a good decision to make here?” … So yeah it just would have been

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good to, to, I suppose just to have some sort of advocacy for the families just so that you can make informed decisions.'-Shanley

Person Centred Care

Carers become experts in the person with dementia as an individual. They act as advocatesand their input into healthcare decisions is essential to achieve true person centred care.

Carers' expertise in their loved one's condition and knowledge of their preferences was routinely overlooked or dismissed by healthcare staff. This left carers feeling disempowered and frustrated.

"there was never any communication with me, when I did try to say something they just ignored me and listened to him"-Jamieson

Early planning and legal context

Proxy healthcare decisions are easier to make when carers have a clear knowledge of the person with dementia's wishes. Encouraging families to have these discussions early in the disease trajectory whilst the person with dementia has capacity to do so facilitates future healthcare decision making.

"Yes, it was very, very difficult. And in hindsight, it would have been a wonderful thing to have had them [conversations] but we didn't."-Shanley

These conversations are difficult and some individuals and families need a lot of support withthem. Explanation of the potential benefits for carers when they are faced with healthcare decisions in the future may help some individuals to address this.

"we tried to talk to mummy about what care she would like further down the line, but she never discuss it... she just wouldn't get into that conversation...and I suppose just when we had to start making those decisions mummy was beyond having any input really in it" -Carter

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The studies were conducted in a range of countries. Each of the included countries have legal provisions which allow individuals to formally nominate one or more individuals to act on their behalf should they lack capacity in future. In the UK, this is known as Lasting Power of Attorney (LPA). The studies in this review showed that some participants were aware of the existence of these legal frameworks. A minority had put them in place. Commonly, carers and people with dementia were resistant to making LPAs.

...I didn't say not at all....I wouldn't do it until much later on I think...I see it as something you hand over when its necessary, not before…-Dening

The mistaken belief that there would be opportunity to set up LPAs once dementia was more advanced led to many reaching a situation where the opportunity was lost.

The only thing that could happen now is Court of Protection . . . because my wife can't sign."-Livingston

One technique that some families used to encourage the person with dementia to make an LPA was to make them for other family members at the same time. This allowed it to be framed as a generic safeguard in case of future eventualities rather than an individual decision necessitated by their inevitable loss of capacity.

"if we did [lasting power of attorney] for both of them [parents], it wouldn't feel like it was just for my mum because she's dementing."-Livingston

Emotional Support

This domain focuses on the emotional impact of making healthcare decisions by proxy and how carers can be supported with this.

Guilt

Making proxy healthcare decisions can be an emotionally fraught process. Health care professionals need to be mindful of the impact that decision making can have on carers.Checking in with carers about the emotional burden and where necessary signposting toservices is important.

"But for a time there, I did feel like I was in charge of ending my mother's life. And I think, if Ihadn't had the psychologist to talk that through, I would have found that really difficult, really difficult "-Shanley

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The burden of guilt was particularly difficult to manage when carers had to make decisions alone or in conflict with family members.

"My brother and sister . . . wanted the drip, antibiotics and the oxygen reinstated . . . and the doctor said 'it will flood her heart' and she died a horrific death . . . but my brother always said,'oh where there's hope' . . . There was no hope." -Livingston

Where decisions were shared, it significantly alleviated the guilt.

"My case manager explains what options there are and what they mean, and I make the decision together with my mother and my sisters. If you work as a team, you make the best choices."-Wolfs

Empathic Guidance

When facing proxy decisions, carers appreciated guidance from health care professionals. It was important however that those decisions were not made unilaterally by professionals.

Recognition and consideration of the needs and concerns of the person with dementia andthe carer were key. This approach is described variously in the included papers, but one paper (Shanley) uses the elegant term empathic guidance.

"And, and I think that was, it was very, it was really helpful to have the medical staff kind of put it in, you know, they didn't beat around the bush … I really felt like they were very supportive and they were looking out for her, and they were looking out for me as well. And so I kind of, I did put a lot of trust in what they said … And they were actually considering her needs at the time and, and so they helped me make that decision."- Shanley

Where health care professionals did not act as empathic guides and failed to demonstrate consideration of the needs of all parties, attempts to engage carers in decision making were often superficial and unsatisfactory.

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"The hospital was no help, and I got aggravated with all of them and finally I had to call a meeting between all the doctors because I could not get anyone to give me any answers."-Truglio Londrigon

Discussion

Summary of main findings

In order to make healthcare decisions for people with dementia, carers require an understanding of the dementia itself and the likely disease trajectory. They also need information about the decision to be made, presented in a form that can be understood andthat can be shared with the person with dementia wherever possible.

Carers may be facilitated in decision making by access to support groups, especially with carers who have faced similar decisions.

Preparation for future healthcare decision making should be facilitated early in the dementia trajectory so that carers feel more confident about the person with dementia's wishes and so that decisions truly are made in their best interests. Health care professionals should encourage families to formalise arrangements for future decision making using the relevant legal structures such as LPAs.

Professionals who recognize the emotional burden of decision making and act as empathicguides for carers making proxy healthcare decisions are highly valued.

Strengths and limitations of the study

This systematic review is novel in that it explores proxy healthcare decision making at earlier stages of the disease trajectory for people with dementia, rather than focusing on end of life care or care home placement decision making. We applied rigorous quality assurance to each study and identified strengths and gaps in the evidence base as well as synthesizing key themes of relevance to practice and policy. We have been guided by the report from the eMERGe reporting guidance which says there are ' no bespoke reporting guidance exists for meta-ethnography...Existing generic guidance for reporting qualitative evidence synthesis pays insufficient attention to reporting the complex synthesis processes of meta-ethnography'.[28] With this in mind, we have used the PRISMA checklist for both abstract and article paper in ensuring there was good practice and

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rigorous, comprehensive analysis,interpretation, and synthesis of the evidence.

A systematic review is by its nature dependent on the quality of the original studies. Of the 15 studies in this review, only one (Livingston) provided detailed demographic data on participants. This study recruited a diverse population in terms of ethnicity, educational background, age and gender. The lack of demographic information in the other studies may suggest a lack of diversity, but certainly suggests a lack of consideration of diversity by the authors. The studies were all conducted in English and excluded the voices of people for whom English is not their first language.

The goal of this review was to focus on the support needs of carers making proxy healthcare decisions for people with dementia. Few of the included papers focused entirely on healthcare decisions from a carer perspective; many dealt with a range of decision types, others included views of other parties such as the person with dementia or the health care professional. As a result, there were often only small sections of the included papers, which yielded relevant quotes.

The thematic synthesis was conducted by 4 academic GPs; VH, CM, CH and SL. Whilst every effort was made to maintain a reflexive approach there will undoubtedly have been shared assumptions and blind spots which could have been mitigated with a more diversere search team.

Comparison with existing literature Information and knowledge about dementia

A lack of knowledge about dementia and the disease trajectory was a common theme throughout this review. A recent systematic review of the public's knowledge and understanding of Alzheimer's disease and dementia found that there was a common misconception that dementia was a normal part of aging. [29] Carers described a need for a post-diagnosis training programme to teach them what to expect. Key within this was the importance of early advanced care planning. Many participants missed the opportunity to have advanced care planning discussions with their loved ones, in part because they did not realise the inevitable downward trajectory of dementia with resultant loss of capacity. Proxy healthcare decisions were much easier to make and accept when carers had a clear understanding of their relative's wishes. Supporting

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carers and people with dementia to have those conversations early facilitates proxy decision making later on.

Shared decision making

Shared decision making is a key component of the NHS long term plan. [4] It is widely accepted that shared decision making facilitates patients to make decisions that are best for them. A 2011 report by The King's Fund [5] states at the heart of decision making 'is the recognition that clinicians and patients bring different but equally important forms of expertise to the decision-making process. In the context of proxy-decision making, the carer should take the role of the second expert, however our review suggests that their expertise is commonly disregarded.

These findings echo the Care Quality Commission's 2016 review of how people are involved in their care, which found limited understanding, recording and monitoring of people's wishes and preferences, inadequate family and carer involvement and insufficient information and explanation of care and support options.[30] Some of the barriers to the implementation of shared decision making identified in the literature include practical factors such as a lack of continuity of care and time constraints. Additionally the challenges involved in building good rapport between both parties have been identified. [31] Some professional groups have proposed solutions such as templates that can be used at various levels of service delivery, [32] while other authors argue that more research is still needed in the area of shared decision making in primary care in order to better conceptualise the barriers and solutions. [33]

Continuity of Care

Many studies have shown the benefits of continuity of care both in terms of patient experience and clinical outcomes. [34-37] The 2018 NICE dementia guidelines recommend that people living with dementia be provided with a single named health or social care professional who is responsible for coordinating their care and that a program of post diagnostic support be offered to carers. [38] In practice, delivery of this support is inconsistent across the UK, with multiple sectors delivering a range of services with the

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potential for duplication or gaps. [39] Our review found similarly inconsistent access to care coordination, but where it was available, carers valued it.

Digital Signposting

The emotional burden of decision making for a person with dementia can be enormous. One of the many consequences of the covid 19 pandemic is the loss of in person support and support networks. [40] Increasing use of on-line support forums offers an alternative for some, but 29% of people aged over 65 in the UK have never used the internet. [41] The movement of services online which has been hugely accelerated by the pandemic risks leaving those without internet access or skills unable to access a wealth of informational, practical and emotional support options.

Implications for future research and clinical practice

The NHS aspires towards making person centred care, including shared decision making, part of routine practice. Research is needed to establish how we can achieve this meaningfully for proxy decision makers, so that both they and the person with dementia have access to the benefits of person centred care. Patients with dementia suffer from inevitable decline in cognition and most will reach a point where they are unable to make healthcare decisions independently. If people with dementia are to benefit from the drive towards patient centred care, it is vital that we consider the support needs of their carers.

Continuity of care can be difficult to achieve in the modern health system, but has undoubted benefits for patients with complex comorbidity. [42] Commissioners should consider how continuity can best be achieved. It should be clear to practitioners and service users who thatsingle point of contact is and how to access necessary support.

In an increasingly digital world, there is potential for widening health inequalities by reducing access to services for those who do not have internet access or computer literacy. A significant minority of older adults does not use the internet and policy makers and commissioners must take measures to ensure that services remain accessible to all, whilst also providing opportunities for people to improve their digital literacy.

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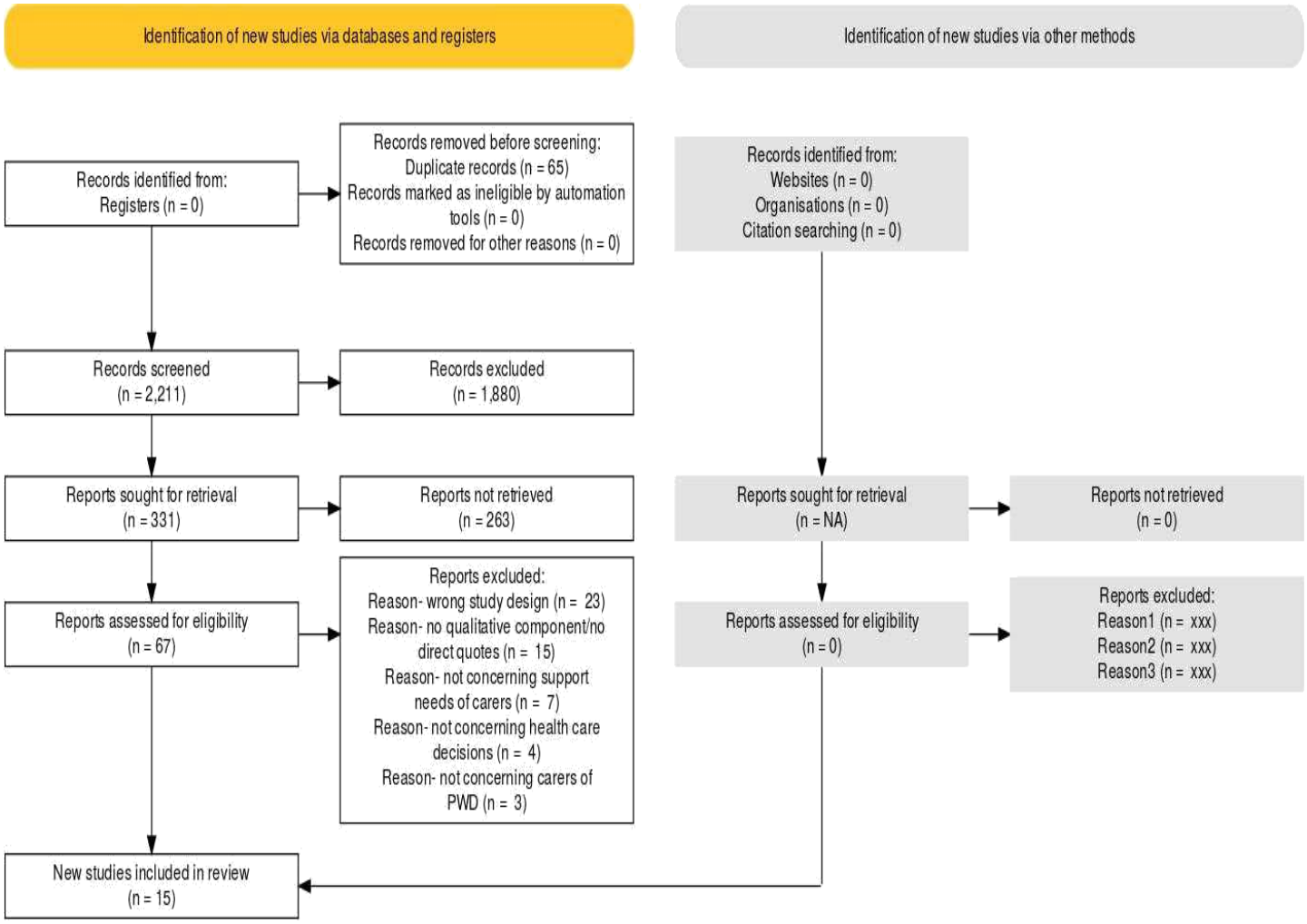
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Data Statement No additional data are available

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Figure 1: PRISMA diagram



Search strategy

MEDLINE via Ovid

<1995-present, limit to English language

1. Carer

2. Caregiver

3. Family

4. 1 or 2 ot 3

5. Decision making

6. Proxy

7. 5 or 6

8. Dementia

9. Alzheimer Disease

10. Cognitive Dysfunction

11. 8 or 9 or 10

12. Qualitative Research

13. 4 AND 7 AND 11 AND 12

14. Remove Duplicates from 14

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Yes
Registration	12	Provide the register name and registration number.	Yes

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title-page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Completed and added as supplementary document
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction, last paragraph, page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	In abstract page 1 and search strategy section of main document page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Eligilbility criteria page 5
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Information sources page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Search strategy page 5
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Study selection and quality appraisal page 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Quality appraisal page 6 and

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PRISMA 2020 Checklist

			data extraction page 6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Data items page 6 and tables 2 and 4
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Quality of reporting page 8 and CASP rating, table 2
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Data extraction page 6
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	PICO table 1
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	CASP rating (table 4)
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Data extraction page 6
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Table 3
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Data extraction page 6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	not used
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Data extraction page 6
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Bias assessment page 7 Data extraction page 6 strengths and limitations page 2
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Synthesis page 8
RESULTS			

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PRISMA 2020 Checklist

Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Data extraction page 6
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	PRISMA diagram figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Strengths and limitations page 13
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	not applicable
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Synthesis page 8
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	no statistical synthesis conducted
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	not applicable
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	not applicable
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Strengths and limitations page 13
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	not applicable
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Summary of findings page 13 and comparison with existing literature page 14
	23b	Discuss any limitations of the evidence included in the review.	Strengths and limitations page 13
	23c	Discuss any limitations of the review processes used.	Strengths and limitations

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PRISMA 2020 Checklist

			page 13
	23d	Discuss implications of the results for practice, policy, and future research.	Implications for future research and clinical practice page 16
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	registration number CRD42020124485
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	https://www.crd.york.ac.uk/PROSPERO
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	not applicable
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 2
Competing interests	26	Declare any competing interests of review authors.	Page 2
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	CASP table

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